



THE STABILITY GAP AS A TRAJECTORY PROBLEM: A BENT DESCENT PATH, ITS OVERSHOOT, AND A SINGLE GATE

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation exposes a failure this view hides: the *stability gap*, a sharp but transient drop in past-task accuracy at the start of every new task. The gap persists even when the optimiser descends the exact joint loss over all tasks, so its cause lies in the optimisation trajectory rather than the objective. This thesis explains the gap on two levels. Learning both tasks has a fixed price: the loss on past tasks must rise to its level at the joint optimum. An envelope-theorem argument shows that a reference path exists along which this rise is monotone, so everything the past-task loss does above that level is avoidable in principle. Steepest descent after an abrupt task switch leaves that path: it bows out of the old task's basin and returns from outside, a deterministic arc that survives exact gradients (level one, the trajectory problem). Real optimisers also overshoot whatever path they follow, pushed by an imbalanced first step and by gradient noise from a finite buffer (level two, the overshoot problem). Written in a unified update rule, every replay method counters these problems with the same tool, a *gate* on the new-task gradient, while the replay gradient restores at full strength. We build and compare the two pure gate designs. The λ -*curriculum* is a scalar feedback gate: it ramps the new-task loss weight up from zero, so the optimiser tracks the moving valley floor of the reference path. *Asymmetric preconditioned experience replay* is a directional feedforward gate: it filters the new-task gradient through past-task curvature, passing it where the past task is flat and braking it where the past task is steep. On rotated MNIST the two levels separate cleanly: removing the overshoot drivers leaves the arc, placing the optimiser on the reference path leaves the overshoot, and doing both removes the gap while the fixed price remains. With momentum the curriculum cuts the gap to a few points of depth at vanilla-ER accuracy. The directional gate lowers the gap monotonically as its filter strengthens, at no plasticity cost but higher compute; at its strongest setting it fails late and abruptly, consistent with blind spots of its sampled curvature metric, and momentum re-inflates its per-step filter. Which gate wins is therefore set by the momentum regime: without momentum the directional gate is better on both stability and accuracy, but with momentum, which realistic training requires, the scalar gate takes over. On CIFAR-10 with a ResNet-18 the curriculum cuts gap depth several-fold below vanilla experience replay at matched final accuracy and negligible compute cost, under two normalisation schemes and on corruptions it was never tuned on.

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1 Introduction

Deep neural networks learn powerful representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [18, 21], is a major hurdle in machine learning. Retraining from scratch solves the issue, but it is too expensive for systems that need to adapt continuously. Continual learning (CL) studies how to learn from a sequence of tasks without that cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [19].

Replay, the family this thesis studies, stores a small buffer of past examples and trains on them alongside the new data. Replay methods reduce long-term forgetting well, but until recently they were evaluated only at task boundaries, which hides what happens within a task. Lange et al. [16] evaluated the network after every update and identified the *stability gap*: a severe but transient collapse in past-task accuracy that occurs immediately when a new task begins, followed by a slow recovery. This matters for two reasons. A deployed system is queried throughout training, so a model caught mid-transition commits errors its final checkpoint never would. And because the model eventually recovers, past knowledge is disrupted rather than lost, which means the disruption itself is the target for repair.

The discovery raised a natural question: would a better approximation of the joint loss over all tasks remove the gap? Hess et al. [11] showed that the gap persists even under incremental joint training, where the model has access to all data from all tasks at every step. A perfect objective still produces transient forgetting. The problem therefore lies in *how* the optimisation proceeds, and this thesis locates it on two levels: the descent path itself is bent, and real optimisers overshoot whatever path they follow.

Two levels. Figure 1.1 visualises the transition as a switch from a learned task (Task A) to a new task (Task B), a conceptual reduction of a high-dimensional surface using the loss-landscape lens of Kao et al. [14]. *Level one* is geometric. Starting at the converged minimum A , steepest descent on the joint loss follows the green curve: because the steepest direction differs from the direction toward the joint minimum $A \cap B$ whenever curvature is anisotropic, the trajectory bows out of Task A’s basin and returns from outside [14], so the Task-A loss rises above its final level and comes back. This arc is deterministic: it appears with exact gradients and vanishing step size, and its signature is a slow, lingering recovery, the *hanging tail*. This increase is an artifact of the specific

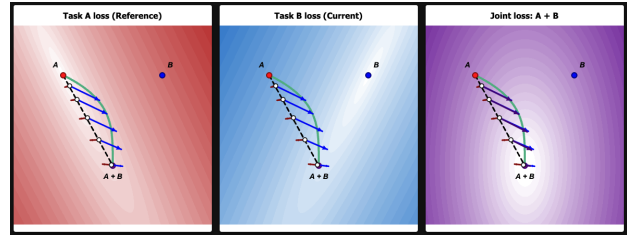


Figure 1.1: 2D optimisation landscape at a task transition. Single-task losses Task A (red) and Task B (blue) are minimised at A and B ; the joint loss (purple) is minimised at $A \cap B$. The steepest-descent trajectory (green curve) illustrates the trajectory problem (Section 3.2), bowing outside A ’s basin into regions of higher loss. In contrast, the dashed black chord sketches the monotone reference path of Section 3.1, which stays inside the basin. Overshoot drivers (Section 3.2) further push the iterate off these paths.

trajectory rather than an unavoidable cost. Reaching the joint optimum requires only that the past-task loss monotonically rise to its higher joint value. Section 3.1 constructs a reference path along which the loss does exactly this, sketched as the dashed chord in Figure 1.1, and the mode-connectivity literature supports the premise that such low-loss routes exist in practice [6, 20]. Everything the accuracy curve does below its settled level and back is therefore avoidable in principle. *Level two* is that real optimisers overshoot whatever path they follow. At the boundary the objective switches in one step from the replay loss to the joint loss, so the target teleports from θ_A^* to θ_{joint}^* while the parameters stand still, leaving the update maximally imbalanced: the replay gradient is near zero while the new-task gradient is at its largest [16]. A finite step on that imbalanced push carries the iterate far out of the old basin, and finite-buffer gradient noise [1] widens and prolongs the excursion. This overshoot acts on the bowed path and the reference alike. Section 3 develops both levels.

One tool: gate the new-task gradient. Sequential replay methods share a single update, a weighted combination of the current-task and replay gradients [22]. Section 3.3 generalises this: every replay method applies a *gate* to the new-task gradient and lets the replay gradient restore at full strength, $d = G g_{\text{cur}} + g_{\text{rep}}$. The gate’s *domain* can be scalar (attenuate the whole push) or directional (attenuate per direction), and its *control* feedback (react to a live signal) or feedforward (commit to a schedule or precomputed filter). This thesis builds and compares the two pure designs. The first is the λ -**curriculum**, a scalar gate. Instead of switching abruptly to the joint loss, it ramps the new-task weight from 0 to 1 over N steps,

$$L_\lambda(t) = L_{\text{replay}} + \lambda(t) L_{\text{current}}, \quad (1.1)$$

so the minimiser of the objective moves along a continuous path from θ_A^* to θ_{joint}^* and the model tracks a moving valley floor. Its adaptive variant sets λ from the ratio of the replay and new-task gradient norms, a live damage signal, which makes it a scalar *feedback* gate. The second is **asymmetric preconditioned experience replay** (asymmetric PER), a directional *feedforward* gate. It filters the new-task gradient through the curvature of the replay loss,

$$d = \delta(F_{\text{rep}} + \delta I)^{-1}g_{\text{cur}} + g_{\text{rep}}, \quad (1.2)$$

where F_{rep} is the replay Fisher information and δ sets the filter strength. The push passes at full gain where the past task is flat and is braked where the past task is steep, while the restoring replay gradient stays untouched.

Contributions. This thesis makes four contributions.

1. **A two-level account of the stability gap.**

The gap separates into a deterministic arc of the descent path and overshoot of the followed path, with the permanent stability–plasticity cost split off by a monotone reference path (Section 3). Controlled experiments confirm the separation: removing the overshoot drivers leaves the arc, fixing the path leaves the overshoot, and doing both removes the gap.

2. **A unified gate view of replay methods,** which places existing methods and the two designs of this thesis in one two-axis taxonomy and turns the taxonomy’s predicted failure modes into testable hypotheses (Section 3.3).

3. **The λ -curriculum,** a scalar feedback gate with an $O(1/N)$ tracking guarantee for its fixed ramp, an adaptive schedule, a λ_{min} plasticity floor, and a characterisation of its interplay with momentum.

4. **Asymmetric PER,** a directional feedforward gate, with a damping sweep that maps its stability–plasticity trade-off and identifies its failure modes: an unexpected late collapse, consistent with blind spots of the sampled curvature metric, and the unravelling of per-step protection under momentum.

Section 2 reviews continual learning, experience replay, the stability gap, and the unified update that underlies the gate view. Section 3 develops the two-level account, constructs the monotone reference path with its tracking bound, and organises replay methods into the gate taxonomy. Section 4 turns the account into four research questions, and Section 5 describes the testbeds, metrics, and conditions that answer them. Section 6 reports the experiments on rotated MNIST and CIFAR-10, and Section 7 weighs the evidence and states the limitations.

2 Background and Related Work

2.1 Continual Learning and Experience Replay

Methods against catastrophic forgetting fall into a few families: replay, parameter and functional regularisation, optimisation-based methods, context-dependent processing, and template-based classification [25]. This thesis focuses on replay, specifically its simplest formulation: Experience Replay (ER) [3]. At each step, ER draws a replay mini-batch from a buffer of stored past examples, combines it with the current-task mini-batch, and takes one optimiser step on the summed loss.

A standard approach to managing this episodic memory is reservoir sampling [26], which ensures that every sample seen so far occupies a buffer slot with equal probability. Under this setup, the buffer remains empty during the first task, meaning ER reduces to plain SGD; consequently, performance gaps and forgetting metrics are evaluated from the second task onward. Despite this simplicity, ER remains highly competitive: a small episodic memory often outperforms purpose-built methods and scales to longer sequences [3]. Furthermore, because the ER update consists of exactly two ingredients, it provides a remarkably clean substrate for isolating and studying interventions at task boundaries.

However, a finite buffer is inherently an imperfect estimator of the past-task distribution. Aljundi et al. [1] frame buffer selection as approximating the feasibility region of the full past data, with reservoir sampling serving as one specific policy within that framework. Consequently, the replay gradient derived from a small buffer will differ in both direction and magnitude from the true gradient of all past data.

2.2 The Stability Gap

Lange et al. [16] identified the stability gap using continual evaluation, measuring past-task accuracy after every update rather than only at task boundaries. In this view a new task begins with substantial but transient forgetting that the model subsequently recovers from, and they quantify it with worst-case metrics, mainly the minimum past-task accuracy reached during training, rather than with end-of-task averages alone. The effect is not confined to one method: they observe it across experience replay, constraint-based replay, knowledge distillation, and parameter regularisation. It also persists well beyond the standard benchmarks, appearing in joint incremental learning of homogeneous tasks [13] and in continual pre-training of large language models [7], and several mitigations have been proposed [9, 23].

Two findings in this literature anchor the account

of Section 3. First, Lange et al. [16] trace the initial drop to a gradient imbalance at the boundary: the new-task gradient dominates the first updates, so the optimiser trades stability for plasticity precisely when the past task is most exposed. Second, the gap is a property of the trajectory rather than the objective. Hess et al. [11] show it persists under training on the exact joint loss with full access to all past data, and Kao et al. [14] demonstrate the geometric mechanism analytically: steepest descent from an old-task minimum leaves the old basin even with perfect joint gradients, because the steepest direction and the direction to the joint minimum disagree. A perfect objective therefore fails to prevent the gap, and the descent geometry itself must be addressed.

2.3 One Update, Many Methods

A second class of replay-adjacent methods constrains the update at the task boundary using information about past tasks. GEM stores past-task gradients and projects the new-task gradient onto the nearest direction that leaves every stored loss unchanged or lower [17]. A-GEM replaces the per-task constraints with a single averaged past-task gradient, reducing the projection to one cheap operation [2]. EWC adds a quadratic penalty around the past optimum weighted by a curvature estimate [15], and natural-gradient approaches precondition the update with past-task curvature so that steps avoid directions the old task depends on [14].

On the surface these methods look unrelated, but Sodhani et al. [22] show the replay-based ones share a single update. Every step combines two gradients, the current task’s and the replay buffer’s, under two scalar weights,

$$\theta \leftarrow \theta - \eta (\alpha_1 \nabla L_{\text{current}} + \alpha_2 \nabla L_{\text{replay}}), \quad (2.1)$$

so a method reduces to a rule for choosing α_1 and α_2 . Plain ER fixes both to one. GEM and A-GEM hold $\alpha_1 = 1$ and raise the replay weight α_2 only when the new-task and stored gradients conflict, by just enough to cancel the interference; A-GEM’s projection is therefore equivalent to re-weighting the two losses. MEGA [8] likewise holds $\alpha_1 = 1$ but sets the replay weight from a loss ratio, $\alpha_2 = L_{\text{replay}}/L_{\text{current}}$, leaning on memory once the new task is already fit.

The same reading applies to the λ -curriculum of this thesis. It fixes $\alpha_2 \equiv 1$ and turns the other knob, ramping the current-task weight $\alpha_1 = \lambda(t)$ up from zero. It belongs to the same family as A-GEM and MEGA; what distinguishes it is which weight it moves and what signal drives the weight. The adaptive variant (Section 3.3) is the mirror image of MEGA: MEGA raises the *replay* weight from a *loss* ratio to recruit memory once the new task is learned, while the adaptive curriculum holds the *new task* back with a *gradient-norm* ratio, letting it in only as the replay

gradient reasserts itself. Section 3.3 generalises Equation (2.1) from scalar weights to operators and organises all of these methods, together with asymmetric PER, along two design axes.

2.4 Low-Loss Paths and Mode Connectivity

Mode connectivity gives empirical backing to the *monotone reference path* introduced in Section 3: the route from the old-task solution θ_A^* to the joint solution θ_{joint}^* along which the past-task loss only ever rises toward its final value, rather than first climbing above it and coming back down as the bowed descent does. For such a path to be usable, the two solutions must be joined by a corridor of low loss in the first place, and this is what the mode-connectivity literature reports. Garipov et al. [6] and Draxler et al. [4] showed that independently trained minima of a deep network are joined by curved paths of near-constant low loss, and Frankle et al. [5] sharpened this to straight paths for networks that share early training. For continual learning specifically, Mirzadeh et al. [20] found that the sequentially trained and the multitask solutions often lie in the same basin, connected by a straight low-loss line. A benign, low-loss route from θ_A^* to θ_{joint}^* should therefore exist in trained networks. Where it does, the stability gap reflects the optimiser straying off that route, and an intervention succeeds by keeping it there.

3 The Stability Gap as a Trajectory Problem

This section reframes the stability gap as a two-level trajectory problem. Learning a new task incurs a permanent stability–plasticity cost; the gap is the transient excess above it. This excess has two sources: a geometric bow in the ideal descent path, and overshoot that real training dynamics add on top. We formalise the split by constructing a monotone reference path, then show that a single design choice, how the update gates the new-task gradient, controls both sources. Two contrasting gates instantiate it: an adaptive scalar curriculum and an asymmetric directional filter.

3.1 The Reference Path and the Tracking Bound

Learning a second task inevitably moves the model. In a quadratic approximation with task losses L_A and L_B and Hessians H_A and H_B , the displacement $d = \theta_{\text{joint}}^* - \theta_A^*$ from the old optimum to the joint minimum permanently raises the replay loss by roughly $\frac{1}{2} d^\top H_A d$. Every method pays this price; it is distinct from the stability gap.

Yet this rise can be strictly monotone. Consider the curriculum objective $L_\lambda = L_{\text{replay}} + \lambda L_{\text{current}}$. As λ grows from 0 to 1, the minimiser traces a *reference path* $\Theta^*(\lambda)$ from θ_A^* to θ_{joint}^* . Under local positive-definiteness, an envelope-theorem argument (Appendix D) guarantees that the replay loss climbs directly to its final level with no excursion above it:

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0 \quad \text{for all } \lambda \in [0, 1]. \quad (3.1)$$

This guarantee is local: it holds only while the combined Hessian $H_A + \lambda H_B$ stays positive-definite along the tracked branch of minima. The condition is automatic near θ_A^* and persists until the new task’s negative curvature overpowers the old task’s; on a deep non-convex backbone it therefore serves as motivation rather than a guarantee.

Figure 3.1 shows this: as λ grows, the loss contours deform continuously from the old-task basin to the joint basin and the minimiser slides along the connected valley floor. Where the condition holds, the replay loss climbs directly to its final level, and any transient excess *above* it, the stability gap, is a preventable deviation from the reference path.

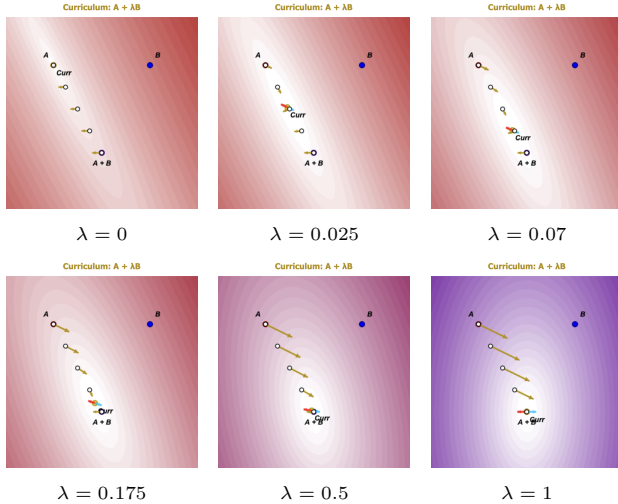


Figure 3.1: Evolution of the curriculum loss $L_\lambda = A + \lambda B$. As λ increases, the contours morph continuously from the old-task basin around A to the joint basin around $A+B$. The current minimiser (*Curr*) slides along the moving valley floor, forming the reference path $\Theta^*(\lambda)$. At the minimiser the replay (red) and current-task (blue) contributions cancel by the first-order condition $\nabla L_{\text{replay}} + \lambda \nabla L_{\text{current}} = 0$; their resultant (golden) vanishes there and, away from the minimiser, points down the valley, so a tracking model rides the moving floor instead of teleporting across the landscape.

A real iterate cannot sit on this path exactly, but it can stay close to it. Introducing the new task gradually over an N -step ramp ($\lambda \approx 1/N$) keeps the iterate tethered to the moving valley floor $\Theta^*(\lambda(t))$ rather

than letting it leap across the landscape; a gradient-flow analysis bounds the resulting lag by $O(1/N)$ (Appendix D). Because the monotone path fixes the endpoint at the permanent level, this caps the highest transient replay loss of the whole transition,

$$\max_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*) + O\left(\frac{1}{N}\right), \quad (3.2)$$

the first term the permanent price, the second a transient that shrinks with ramp length. A slow ramp is precisely what the λ -curriculum of this thesis implements.

3.2 The Two-Level Gap: Bowing and Overshoot

Real trajectories deviate from the reference path on two distinct levels.

Level One (The Bow): Even under idealised conditions (exact gradients, vanishing step size), the steepest-descent direction of the abruptly assembled joint loss misaligns with the direction toward the joint minimum whenever the loss curves more sharply in some directions than in others. The trajectory bows outward, leaving the old task’s basin before returning [14]. This deterministic out-and-back arc forms the structural core of the gap (Figure 1.1, green curve).

Level Two (Overshoot): Real training dynamics inflate this arc. At a sudden task boundary, the minimum teleports from θ_A^* to θ_{joint}^* , leaving the iterate exposed to a large new-task gradient while the replay gradient is near zero [16]. The acute driver is this imbalance met by a finite step size: the first discrete steps ride the unopposed push far out of the old basin, creating the sharp first-step drop. A chronic driver, estimator variance from finite buffer sampling (Section 2.1), injects noise into the restoring gradient at every step; it slows the recovery tail and can cause sudden drops long after the boundary.

The two levels shape the gap’s two dimensions: depth records the overshoot, whether the acute boundary spike or a late noise-driven drop, while the hanging tail is the Level-One arc plus the slow, noisy recovery the chronic driver leaves behind. This yields the falsifiable core, formalised as RQ1 in Section 4: removing the overshoot drivers should flatten the drop and leave the arc, placing the optimiser on the reference path should remove the arc and leave the overshoot, and doing both should close the gap while the permanent cost of Section 3.1 remains.

3.3 Two Gates: Attenuating the New-Task Gradient

Equation (2.1) shows that replay methods differ mainly in how they weight the replay and current-task gradients. Because a common scale of the two weights is absorbed by the learning rate, only their

ratio matters, so any such method can be rewritten with the replay weight normalised to one. The replay gradient is the only restoring force pulling the iterate back to the reference path, so a protective method must attenuate the new-task push, the source of the bow and overshoot, while leaving the restoring gradient intact. Generalising the current-task weight to an operator G yields a one-sided *gate*:

$$d = G g_{\text{cur}} + g_{\text{rep}}. \quad (3.3)$$

The curriculum of Section 3.1 realises this gate along one axis, *time*: it introduces the new task slowly, so the push is never large enough at once to knock the iterate off the reference path. A complementary axis is *direction*. At any single step only part of the push is harmful, because the past task is flat in most directions and steeply curved in a few, and only motion along the steep directions raises the replay loss. Preconditioning the push by past-task curvature exploits this, passing it at full gain where the past task is flat and shrinking it where the past task is steep, so retention costs little plasticity. This natural-gradient idea underlies Natural Continual Learning [14] and, in a replay setting, Online Curvature-Aware Replay [24]. Both, however, precondition the *entire* update, so the same metric that brakes the harmful push also brakes the restoring replay gradient, protecting the past task only by slowing the return to it. Asymmetric PER keeps the gate one-sided: it filters the new-task gradient alone and leaves the restoring gradient at full strength. We include the symmetric preconditioner, which damps both, as a control (Appendix C.5).

As detailed in Table 3.1, gates vary by their *domain*, scalar or directional, and their *control*: a feedback gate reacts to a live damage signal, while a feedforward gate commits in advance to a schedule or a precomputed filter. This thesis explores the two pure corners of this design space; a fixed linear ramp, the curriculum’s feedforward variant, serves as an ablation:

Adaptive λ -curriculum (Scalar Feedback): A scheduling parameter $\lambda(t)$ sets the new-task weight from the ratio of replay to new-task gradient norms, a live damage signal that fires whichever direction harm arrives from. By scaling the loss itself, it deforms the objective into the homotopy of Section 3.1, so the optimiser’s natural descent follows the reference path.

Asymmetric PER (Directional Feedforward): The gate $G = \delta(F_{\text{rep}} + \delta I)^{-1}$ is a fixed filter set by the replay Fisher, with the damping δ its single knob: as $\delta \rightarrow 0$ the push concentrates in the directions the past task is flat in. Committed in advance, it is feedforward.

The taxonomy yields the hypotheses that Section 4 formalises as research questions. The scalar feedback

Method	Gate G	Domain	Control
Vanilla ER	I	—	—
GEM / A-GEM	reweight on conflict	scalar	feedback
MEGA	loss ratio	scalar	feedback
Balancing	norm matching	scalar	feedback
Linear curric.	$\lambda(t) I$, clock	scalar	feedforward
Adaptive curric.	$\lambda(t) I$, norm ratio	scalar	feedback
Asym. PER	$\delta(F_{\text{rep}} + \delta I)^{-1}$	directional	feedforward

Table 3.1: Replay methods as gates on the new-task gradient, Equation (3.3). Balancing is the gradient-norm-matching control D3 of Section 5.2.1. The two methods of this thesis occupy the pure corners: a scalar feedback gate and a directional feedforward gate.

gate holds back the whole push and so pays a plasticity cost a separate accelerator must repay. The directional feedforward gate should protect the past task at little plasticity cost. The corners also differ in computational cost: the scalar gate adds only a gradient-norm ratio per step, while the directional gate solves a linear system in the replay Fisher at every step, by conjugate gradients over Fisher-vector products (Appendix A.4).

Momentum is the natural accelerator, and it should discriminate between the corners. On vanilla ER it should leave the depth unchanged, because the unopposed first step at the boundary carries no accumulated velocity, while its velocity average smooths the noisy replay gradient and damps the chronic driver. The curriculum attenuates *at source*: with $\lambda \approx 0$ early there is nothing to accumulate, so momentum’s averaging and acceleration compose with the gate and repay its plasticity debt. The preconditioner attenuates each *step* while the gradient keeps its size, so velocity accumulated across steps can rebuild the displacement the filter removed, and momentum should partly undo the protection. How a gate composes with momentum is therefore set by where it attenuates, at the source or at the step, and a head-to-head comparison of the corners should depend on the momentum regime rather than trade along a fixed stability–plasticity frontier.

4 Research Questions

The two-level account and tracking bound of Section 3 make falsifiable predictions about the gap and each gate. The experiments are organised around four questions. The first three are answered on rotated MNIST with a small MLP, where the gap can be probed step by step across many conditions and seeds; the fourth asks whether those answers survive a deeper backbone and a stronger task shift.

RQ1 — The account. Does the gap separate into a deterministic arc of the descent path and overshoot of the followed path, such that removing the over-

shoot drivers leaves the arc, placing the optimiser on the reference path leaves the overshoot, and doing both removes the gap while the permanent stability-plasticity cost stays visible as reduced plasticity?

RQ2 — The scalar feedback gate. Does the λ -curriculum reduce the gap by timing the new-task push, and how do its ingredients divide the work? Four sub-questions probe the design:

RQ2a (linear ramp). Does gap depth fall as the ramp length N grows, consistent with the $O(1/N)$ bound?

RQ2b (adaptive schedule). Does the gradient-norm feedback signal match the best fixed ramp while removing the knob N ?

RQ2c ($\lambda_{\min}$ floor). Does a small floor $\lambda_{\min} > 0$ recover early plasticity at a modest depth cost?

RQ2d (momentum). Does momentum compose with the curriculum, closing the stochastic residual and restoring final accuracy, while leaving the depth of vanilla ER unchanged?

RQ3 — The directional feedforward gate. Does asymmetric PER reduce the gap as its filter strengthens, at what plasticity cost, and where does a feedforward gate fail? The account predicts one failure axis, momentum, whose accumulated velocity rebuilds the push the filter shrinks one step at a time.

RQ4 — Generalisation. Do the findings carry to a deeper backbone (ResNet-18) and a stronger task shift (CIFAR-10 domain corruption): does each gate still cut the gap at matched final accuracy, and do the momentum signatures of RQ2d and RQ3 reproduce there in a momentum cross?

5 Methods

This section describes the empirical testbeds (Section 5.1), the metrics that summarise each run (Section 5.1.2), and the experimental blocks (Section 5.2) that answer the research questions of Section 4.

5.1 Experimental Setup

5.1.1 Testbeds

Two testbeds of different scale carry the study. A small, fast one (rotated MNIST on a two-layer MLP) runs the dense mechanistic sweeps: the driver block, the curriculum sweep, the asymmetric-PER damping sweep, and the momentum cross. A larger one (domain-shift CIFAR-10 on a CIFAR-adapted ResNet-18) tests whether the findings survive a deeper backbone and a stronger task shift, the regime where

the local condition behind the bound of Section 3.1 weakens. Both hold the label space and task structure fixed and shift only the input distribution, and both use two tasks in their main blocks, so there is exactly one transition and one instance of the stability gap; a five-task rot-MNIST sequence and a three-task CIFAR sequence serve as multi-boundary checks.

On rot-MNIST, following Lange et al. [16], T_0 presents the digits at -90° and T_1 applies a further 90° rotation, each task using the standard 60,000 training and 10,000 test images. On CIFAR-10, T_0 is clean and T_1 applies Gaussian-noise corruption at severity 3 [10]; the ten classes are shared, so only the input distribution shifts. A three-task variant ([none, Gaussian, shot]) and two alternative corruptions appear in the generalisation block. The rot-MNIST MLP is cheap enough to sweep across many conditions and seeds and is where the local condition of Section 3.1 is most plausible; the ResNet-18 is the deeper, more realistic counterpart. rot-MNIST runs online (one epoch per task), standard for stability-gap analysis; the ResNet needs several passes to converge before the transition, so the CIFAR block uses ten epochs per task. Optimiser, learning rate, and batch size are common across benchmarks. Full backbone and protocol specifications are in Appendix A.2 (Tables A.1–A.3).

Throughout training every task’s test accuracy is evaluated every 10 steps; at each transition the cadence rises to every step, so the dip is recorded at full resolution (the whole online epoch on rot-MNIST, a 250-step window on CIFAR). A stability-gap tracker snapshots each past task’s pre-switch accuracy and records the resulting (step, accuracy) series. For the driver block only, a gradient tracker additionally records the fidelity of the replay-buffer gradient against the exact past-task gradient, at the same cadence; each such step costs a full pass over the past-task training set, which is why it is not run elsewhere.

5.1.2 Metrics

Each metric is reported as the seed mean $\pm$ sample standard deviation; complete definitions are in Appendix B. The gap definition of Section 3, the transient excess below the level a task settles at, gives two primary metrics. With a_j^{pre} the pre-switch baseline on past task j and $a_j(t)$ its accuracy at step t , **gap_depth** is the maximum instantaneous excursion $\max_j(a_j^{\text{pre}} - \min_t a_j(t))$, which Section 3.2 predicts to be overshoot-dominated. The second, **gap_area_end**, integrates the shortfall below the settled level: with reference $b_j = \min(a_j^{\text{pre}}, \bar{a}_j)$, where $\bar{a}_j$ is the accuracy the task eventually reaches, it sums $\max(0, b_j - a_j(t))$ over the new task’s training steps and across past tasks. Anchoring to the recovered level implements the split of Section 3.1: a permanent step down to a lower plateau contributes to the standard forget-

ting metrics and near-zero to the area, while a true dip-and-recover registers in full. One consequence, revisited when reading the tables, is that two methods settling at different levels are measured against different lines, so a better-recovering method can carry the *larger* area; a positive area marks transient disruption, not net cost, and we read it with final accuracy. The arc of Section 3.2 and the slow part of the noisy recovery live here; sharp noise-driven drops register in depth instead. A third metric, **recovery_steps**, and the gradient-fidelity diagnostics are defined in Appendix B.

To place the gap in the standard frame we report the continual-evaluation metrics of Lange et al. [16] from the task-boundary accuracy matrix: average final accuracy (ACC), which checks that a gate lowers the gap at preserved accuracy; worst-case retention (min-ACC, WC-ACC); and short-horizon plasticity (WP_{100}). FORG and the windowed WF_w/WP_w family are logged for completeness. Conditions are compared with the paired Wilcoxon signed-rank test on seed-paired differences, preferred over the t -test because five paired seeds cannot support a normality assumption on bounded metrics: the gap metrics floor at zero, where the strongest conditions cluster, and accuracy sits near its ceiling, so the seed differences are skewed rather than Gaussian. With five paired seeds the smallest two-sided p -value is 0.0625, reached when all five differences share a sign; we read $p = 0.0625$ as complete directional agreement and let effect sizes separate large results from marginal ones.

5.2 Experiments

Each block maps to one or more research questions and tests the corresponding predictions of Section 3. All conditions use five seeds and the protocol above unless noted.

5.2.1 Overshoot Drivers (RQ1)

The driver block answers the overshoot side of RQ1: each condition switches off one of the drivers of Section 3.2 by construction and observes what remains. The complementary path-side intervention is the curriculum, whose full-buffer diagnostic below completes the RQ1 cross. Table 5.1 lists the four conditions. Balancing (D3, D4) gives g_{new} and g_{replay} equal magnitude from the first step, switching off the boundary imbalance: an unbalanced step at θ_0^* (the converged T_0 optimum, θ_A^* of Section 3) is driven almost entirely by g_{new} , a balanced step carries the replay direction at equal weight. The full-data buffer (D2, D4) replaces vanilla ER’s noisy 256-sample replay gradient with the exact empirical past-task gradient, switching off the estimator variance (mechanics in Appendix A.3). Because the drivers interact (Section 3.2), we read the conditions as interventions, not an additive decomposition: depth differences measure how much of the

drop each driver carries in that context, while the area and per-step shape carry the level-one signal: with variance off (D2, D4), any dip-and-recover bend that remains is deterministic, the arc’s fingerprint.

5.2.2 The λ -Curriculum Sweep (RQ2, RQ1)

The curriculum sweep tests the scalar feedback gate one ingredient at a time: the $O(1/N)$ prediction of Equation (3.2) (RQ2a), the adaptive schedule (RQ2b), and the $\lambda_{\min}$ floor (RQ2c); a full-buffer diagnostic (C8) then returns to RQ1, testing the claim that the curriculum removes the arc and leaves only stochastic overshoot. All conditions run on top of standard ER (1k reservoir, unbalanced gradients), so the schedule is the only changed variable (Table 5.2). The adaptive schedule (C4) is the feedback variant: it sets λ to a clipped exponential moving average of the gradient-norm ratio $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$. At θ_0^* the replay gradient is near zero by stationarity, so r and λ start near 0; as the iterate moves, $\|g_{\text{replay}}\|$ grows and $\|g_{\text{new}}\|$ shrinks, so $\lambda \rightarrow 1$ at the pace the damage signal dictates, with no ramp length to choose. The floor $\lambda_{\min}$ (C5–C7) keeps the schedule off $\lambda \approx 0$ during the first steps and buys back early plasticity.

C8 combines the practical curriculum configuration (C7) with the full 60k buffer of D2/D4, so the path is fixed and the variance driver is switched off at source. This makes it a double probe. First, it tests what C7’s remaining momentum-0 transient is made of: the curriculum has already removed the arc, so if the residual is stochastic overshoot from buffer noise, the exact replay gradient should remove it too and C8 should show no gap even without momentum. Second, comparing C8 with and without momentum (C8 vs. $C8_M$) separates the two roles momentum could play for the curriculum. The full buffer already removes the gradient noise, so C8 has no variance left for momentum to reduce; whatever momentum still improves on top of C8 must come from its other role, accelerating new-task learning. C8 stores the whole past task and so breaks the limited-memory premise; it is a diagnostic only, and C7 remains the practical configuration.

5.2.3 Momentum Cross (RQ2d, RQ3)

Every rot-MNIST condition of the driver D-series and the curriculum C-series, and the asymmetric-PER sweep legs P1–P2 at the settings noted in Table 5.3, is run at momentum 0.0 and 0.9; momentum-on runs carry an M subscript (e.g. $D1_M$). The cross tests the composability prediction of Section 3.3: whether momentum helps or hurts a method’s gap should depend on where the method attenuates the push. A five-task rot-MNIST sequence (rotations $[0, 30, 60, 90, 120]^\circ$, four transitions, five seeds) repeats the three-way comparison of vanilla ER, the practical curriculum, and asymmetric PER at both momentum settings, as a multi-boundary check (Appendix C.9).

Label	Condition	Replay buffer	Gradient handling		Driver removed by construction
D1	Vanilla ER	1k reservoir	unbalanced		none (all drivers active)
D2	Full-data ER	60k (all past)	unbalanced		estimator variance (exact replay gradient)
D3	Balanced ER	1k reservoir	unit-norm	balanced	boundary imbalance
D4	Full-data + balanced	60k	unit-norm	balanced	imbalance and variance

Table 5.1: Driver-removal conditions. Each switches off one or two overshoot drivers relative to vanilla ER; what all four leave in place is the descent path itself, so the arc should survive throughout.

Label	Condition	$\lambda(t)$ schedule	Hyperparameter	Tests
—	Vanilla ER (baseline)	$\lambda \equiv 1$ from step 1 (abrupt switch)	—	reference
C1	Linear $N=50$	$\text{clip}(t/50, 0, 1)$	$N = 50$	RQ2a
C2	Linear $N=100$	$\text{clip}(t/100, 0, 1)$	$N = 100$	RQ2a
C3	Linear $N=200$	$\text{clip}(t/200, 0, 1)$	$N = 200$	RQ2a
C4	Adaptive	$\text{clip}(\text{EMA}_\alpha(r), 0, 1)$	$\alpha = 0.05$	RQ2b
C5	Adaptive + $\lambda_{\min}=0.05$	$\max(\lambda_{\min}, \text{clip}(\text{EMA}_\alpha(r), 0, 1))$	$\alpha = 0.05, \lambda_{\min}=0.05$	RQ2c
C6	Adaptive + $\lambda_{\min}=0.10$	as C5	$\lambda_{\min}=0.10$	RQ2c
C7	Adaptive + $\lambda_{\min}=0.20$	as C5	$\lambda_{\min}=0.20$	RQ2c
C8	C7 + full 60k buffer	as C7, exact replay gradient	diagnostic	RQ1, RQ2d

Table 5.2: λ -curriculum conditions, all applied on top of standard ER. EMA_α is an exponential moving average with smoothing α , and $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ is the gradient-norm ratio. Each condition is run at momentum 0.0 and 0.9.

Label	Condition	Replay buffer	Damping δ	Purpose
P1	Standard leg	1k reservoir	1.0, 0.3, 0.1, 0.03	practical regime, matched to the curriculum (RQ3)
P2	Full-buffer leg	60k (exact g_{rep})	1.0, 0.3, 0.1, 0.03	variance driver off; residual area is the arc (RQ3, RQ1)
P3	Solver control	60k	0.03	60 CG iterations instead of 25; rules out solve artefacts
P4	Symmetric control	1k reservoir	1.0, 0.3, 0.1, 0.03	preconditions $g_{\text{cur}} + g_{\text{rep}}$; ablates the one-sided gate

Table 5.3: Asymmetric-PER conditions. Both sweep legs run at momentum 0.0; momentum-0.9 counterparts cover $\delta \leq 0.3$ on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg. The two controls run at momentum 0, where the effects they check for arise. The filter metric is always the replay Fisher F_{rep} , estimated on a sampled batch (Appendix A.4).

5.2.4 Asymmetric-PER δ Sweep (RQ3)

The damping sweep tests the directional feedforward gate: does the gap shrink as the filter strengthens, at what plasticity cost, and where does the gate fail? Asymmetric PER implements Equation (1.2), filtering the new-task push through the replay Fisher F_{rep} [12]. Normalising by δ keeps flat directions at unit gain: along an eigendirection of curvature ρ the gain is $\delta/(\rho + \delta)$, so directions the past task is flat in pass through whole and steep ones are shrunk toward zero. The damping δ is the single knob: at $\delta \rightarrow \infty$ plain ER returns, while as $\delta \rightarrow 0$ the push concentrates in replay-flat directions with the restoring gradient g_{rep} at full strength throughout. The push is applied matrix-free by conjugate gradients over Fisher-vector products (Appendix A.4).

Table 5.3 lays out the sweep: $\delta \in \{1.0, 0.3, 0.1, 0.03\}$

crossed with two buffer legs, each at both momentum settings. The *standard* leg (1k reservoir) matches the curriculum’s practical regime, while the *full-buffer* leg makes the replay gradient exact, switching off the variance driver as in D2/D4, so the remaining area measures the deterministic arc and the sweep reads out directly whether the gate shrinks the arc itself. Two controls guard the reading: a solver control reruns the strongest filter at 60 CG iterations to separate gate properties from an under-converged solve (Appendix A.4), and a symmetric control preconditions the summed gradient by the same replay Fisher, isolating the effect of leaving the restoring gradient untouched (Appendix C.5).

5.2.5 Generalisation: CIFAR-10 (RQ4)

The CIFAR-10 block tests whether both gates and the momentum signatures carry to a deeper backbone and a stronger task shift. All CIFAR conditions use the ResNet-18 protocol of Table A.3 (ten epochs per task, buffer 5,000). The headline conditions run at momentum 0.9 (Table 5.4, top), each method crossed with BatchNorm and GroupNorm because the two respond differently at the switch: the input shift disrupts BatchNorm’s running statistics, which must re-adapt during the first steps, while GroupNorm uses no cross-batch statistics, so its transient reflects optimisation dynamics alone. The curriculum runs as the adaptive schedule at $\lambda_{\min} = 0.20$; asymmetric PER runs at $\delta = 0.1$, a strong rot-MNIST setting clear of the blind-spot regime. Both carry their rot-MNIST configurations without a CIFAR-specific sweep, so they meet the deeper backbone on equal footing. A momentum cross (Table 5.4, bottom) adds momentum-0 runs of all three methods at GroupNorm, repeating the three-way composability test where the transient is free of normalisation-statistics effects; because momentum also drives convergence on this backbone, accuracy is compared within a momentum setting. A further generalisation block pairs the practical curriculum against a paired vanilla-ER control on shifts it was never tuned on (a two-task shot-noise shift, a two-task contrast shift, and a three-task [none, Gaussian, shot] sequence, each at both normalisations over three seeds) to test whether the result depends on the particular corruption or on having a single transition.

Label	Method	Norm	μ
G1	Vanilla ER	BatchNorm	0.9
G2	Adaptive $\lambda_{\min}=0.20$	BatchNorm	0.9
G3	Asym. PER $\delta=0.1$	BatchNorm	0.9
G4	Vanilla ER	GroupNorm	0.9
G5	Adaptive $\lambda_{\min}=0.20$	GroupNorm	0.9
G6	Asym. PER $\delta=0.1$	GroupNorm	0.9
G7	Vanilla ER	GroupNorm	0.0
G8	Adaptive $\lambda_{\min}=0.20$	GroupNorm	0.0
G9	Asym. PER $\delta=0.1$	GroupNorm	0.0

Table 5.4: CIFAR-10 conditions (two-task Gaussian-noise shift, buffer 5,000, five seeds). Top: the headline block at momentum 0.9. Bottom: the momentum cross, pairing momentum-0 runs of all three methods against their momentum-0.9 counterparts at GroupNorm. The generalisation block (novel corruptions and the three-task sequence, three seeds) is described in the text and reported in Appendix C.7.

6 Results

The results follow the research questions: the two-level account (Section 6.1, RQ1), the scalar feedback gate

(Section 6.2, RQ2), the directional feedforward gate (Section 6.3, RQ3), a synthesis of the two momentum crosses (Section 6.4), and the CIFAR-10 generalisation of both gates (Section 6.5, RQ4).

Reading the tables. Depth is in percentage points (pp): a depth of 19.5 means the worst past-task evaluation falls 19.5 points below its pre-switch baseline. Section 3.2 predicts depth to be overshoot-dominated, and both overshoot drivers express in it: the acute boundary spike, and the buffer’s noise, whose sudden drops can set the worst evaluation long after the switch. The end-referenced area `gap_area_end` (accuracy-steps) integrates the excursion below the level each task settles at, so a permanent drop scores near zero and contributes instead to forgetting. Because each condition’s area is taken against its own settled level (Section 5.1.2), area is comparable only between conditions that recover to the same level, so we read cross-condition area together with final accuracy and the per-step curves. Values are mean $\pm$ standard deviation over five seeds (three for the CIFAR generalisation block). Following Section 5.1.2, the two-sided Wilcoxon test floors at $p = 0.0625$ when all five seed differences share a sign, so we quote a p -value only where the seeds disagree and otherwise report complete directional agreement. The full metric set for every condition is in Appendix C.

6.1 The Two Levels, Separated (RQ1)

RQ1 asks whether the gap separates into a deterministic arc of the descent path and overshoot of whatever path is followed, such that removing the overshoot drivers and fixing the path are separately insufficient but jointly sufficient. Crossing the two interventions at momentum 0 gives the 2×2 of Figure 6.1: colour encodes the path (red: bowed steepest descent, no gate; blue: the curriculum’s valley-floor path, which tracks the reference path of Section 3.1), linestyle the overshoot drivers (solid: present; dashed: removed), so each curve is one cell of the cross.

Removing the drivers leaves the arc. Following the driver ladder (D1–D4, full metrics in Appendix C.1, Table C.1), unit-norm balancing (D3) removes the boundary imbalance and takes 13.8 pp off the depth ($19.5 \rightarrow 5.7$), by far the largest single drop; swapping the reservoir for the exact past-task gradient (D4) removes the estimator variance and takes off a further 3.8 pp ($5.7 \rightarrow 1.9$). Gradient-fidelity diagnostics confirm the driver each condition switches off: the finite-buffer D1 and D3 reach only $\text{c\ddot{o}s}(g_{\text{replay}}, g_{\text{true}}) \approx 0.71$ and 0.66 with per-step minima near zero, while the full-data D2 and D4 sit at 1.000 by construction (Appendix C.3). One nuance keeps the ladder honest: balancing is not a pure imbalance intervention, since fixing both gradients to

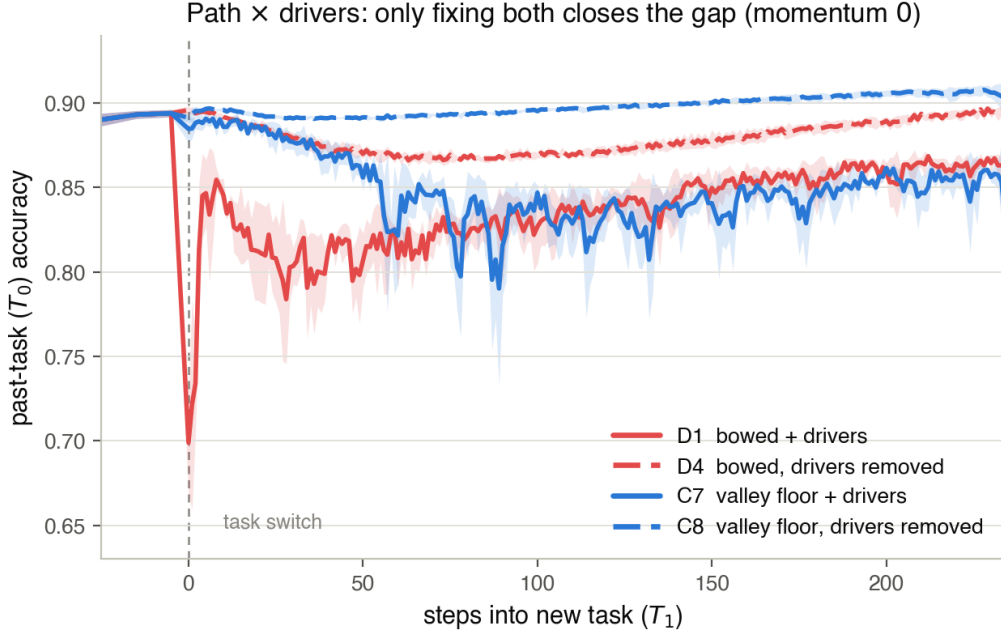


Figure 6.1: The RQ1 2×2 as per-step curves: past-task (T_0) accuracy across the rot-MNIST switch at momentum 0; seed mean with $\pm$ std bands, $x=0$ the task switch. Colour crosses the path (red: bowed steepest descent, no gate; blue: the curriculum’s valley-floor path), linestyle the overshoot drivers (solid: present; dashed: removed by balancing and the exact full-buffer replay gradient). Cell metrics as `gap_depth (pp) / gap_area_end / ACC`: D1 19.5/6.3/0.873, D4 1.9/3.2/0.869, C7 13.9/2.7/0.837, C8 0.2/0.05/0.818. Removing the drivers alone (D4) leaves the smooth deterministic arc; fixing the path alone (C7) leaves the overshoot, arriving as sudden noise-driven drops well after the switch; doing both (C8) holds T_0 flat through the switch, while the permanent stability–plasticity cost stays visible as the lowest final accuracy of the four.

unit norm also strips the magnitude component of the buffer noise and caps how far any single bad replay estimate can move the iterate, so part of variance’s *expression* in depth leaves with the imbalance. The driver itself stays live — the direction is no less noisy (the cosine, if anything, worsens) and D3’s curve keeps its jitter — which is why we read the ladder as interventions, not an additive decomposition (Section 5.2.1). What no driver removal touches is the descent path, and the area shows it: with the spike gone at D4 (only 1.9 pp) a clear bend remains, `gap_area_end` = 3.2. Because the full-data conditions carry zero gradient noise, that area cannot be stochastic residual; it is the deterministic arc of Section 3.2. Figure 6.1 shows it directly: with the drivers removed, D4 (dashed red) traces a smooth dip-and-recover with no jitter, the fingerprint the account attaches to a bowed path rather than a spike. (That D2’s area, 8.1, exceeds D1’s 6.3 is the end-referenced metric at work, not a larger arc: D2 recovers to a higher level, so more of its excursion counts as transient; Section 5.1.2.) The full ladder, including the intermediate D2 and D3, is plotted in Appendix C.10 (Figure C.1).

Fixing the path leaves the overshoot. The other row holds the drivers in place and fixes the path with the curriculum. At momentum 0 the prac-

tical curriculum (C7, solid blue) still shows 13.9 pp of depth, and Figure 6.1 shows where it comes from: the boundary spike is gone, and the depth arrives instead as sudden noise-driven drops well after the switch, the chronic driver expressing in depth as Section 3.2 predicts. With the arc removed, essentially all of C7’s gap is overshoot. That this residual is estimator variance rather than anything structural is settled by C8, which adds the exact full buffer to C7 so that the path is fixed *and* the variance driver is switched off at source. The gap then collapses to 0.2 ± 0.3 pp of depth and 0.05 of area, the smallest depth of any condition in the study, and the C7→C8 area drop ($2.7 \rightarrow 0.05$) holds across all five seeds. Overshoot is real and untouched by the path fix; the arc is real and untouched by the driver removal; only doing both closes the gap.

The permanent cost stays on the ledger. Closing the transient does not make the transition free. C8’s final accuracy is 0.818, the lowest of the four cells and 0.055 below vanilla ER, because holding the new-task push down at momentum 0 slows the new task without an accelerator to repay it. This is the permanent stability–plasticity cost the account predicts to remain once the transient is removed; Section 6.2 shows momentum paying it back. A second, independent line of evidence that the arc is deterministic ge-

ometry comes from the directional gate (Section 6.3): on its full-buffer legs, where the variance driver is off and the area *is* the arc, the area shrinks monotonically as the filter strengthens ($2.3 \rightarrow 1.0 \rightarrow 0.4$ for $\delta = 1.0 \rightarrow 0.3 \rightarrow 0.1$). The arc responds to path-level intervention and to nothing else, whether the intervention times the push (the curriculum) or filters its direction (the gate).

6.2 The Scalar Feedback Gate (RQ2)

Table 6.1 reports the curriculum sweep at both momentum settings against vanilla ER. The momentum-off leg tests the schedule in isolation; the momentum-on leg is the practically relevant regime.

Lengthening the ramp reduces gap depth (RQ2a–c). At momentum 0, mean gap depth falls monotonically as the linear ramp lengthens, as the $O(1/N)$ bound of Equation (3.2) predicts (RQ2a), though successive lengths lie within the seed spread and the reduction is bought at a real accuracy cost: by $N = 200$ ACC has fallen from vanilla’s 0.873 to 0.828 and the tail has stretched, **recovery_steps** growing from 36 at $N=50$ to 133 at $N=200$, against vanilla’s ≈ 4 (Appendix C). The depth that remains, still in the mid-teens, is the untouched chronic driver: the schedule times the push but leaves the finite-buffer replay estimate exactly as noisy as vanilla’s, so the worst evaluation keeps arriving as a late noise-driven drop, a floor no ramp length can pass and only the exact-gradient diagnostic removes (C8, Section 6.1). The adaptive schedule (C4) reaches 12.8 pp, below the best linear ramp, without a ramp length to set (RQ2b; C3→C4 agrees in direction but stays within noise, $p = 0.44$). Raising the floor $\lambda_{\min}$ to 0.20 (C7) shifts the schedule in the predicted Pareto direction only weakly at momentum 0, because early T_1 progress is then set by the large deterministic boundary gradient the small floor cannot redirect (RQ2c). The momentum-0 curves of Figure 6.2 (dashed) show the trade directly: the curriculum reaches a shallower past-task dip than vanilla (a) but pays for it in slower early T_1 learning (b).

Momentum pays the plasticity debt, and closes the gap (RQ2d). The momentum-on leg is the scalar gate’s intended regime, and it acts through momentum’s two roles at once. Adding momentum to the adaptive schedule drops depth to 2.5 pp (C4→C4_M) and to 3.7 pp at the $\lambda_{\min} = 0.20$ floor (C7_M), collapses the area to near zero, and restores accuracy to vanilla ER’s level (Table 6.1); every one of these contrasts holds across all five seeds. The depth and area collapse is variance reduction: momentum averages the noisy finite-buffer replay gradient across accumulated velocity, damping the chronic driver that feeds the residual dip and tail once the schedule has tempered the

acute boundary push at source. The accuracy restoration is the second, separate role: acceleration, driving the new task hard enough to repay the plasticity the schedule borrows. The floor whose effect was weak at momentum 0 becomes a clean Pareto knob once momentum is active, and what it trades is not transient stability: in this regime the past task no longer dips and recovers but settles at a marginally lower plateau (area ≈ 0), so depth here reads the permanent stability–plasticity split itself, where between the two tasks the shared parameters come to rest. Raising the floor from 0 to 0.20 (C4_M → C7_M) shifts that split toward the new task, and the final per-task accuracies (from the logged accuracy matrix, Appendix B) show both ends of the trade: T_0 settles at 0.921 instead of 0.936 (the step depth reads as $2.5 \rightarrow 3.7$ pp, FORG as $0.019 \rightarrow 0.035$) while T_1 finishes at 0.941 instead of 0.899 — 4.2 pp of new-task accuracy for 1.5 pp of old, both trends across all five seeds — with the gain already visible at short horizon (WP₁₀₀; Appendix C) because a higher floor lets T_1 train from the first post-switch step. C7_M is the configuration carried to CIFAR.

That momentum here closes a gap it left untouched under vanilla ER is the composability that Section 3.3 predicts for a gate that attenuates at source, and a full-buffer diagnostic separates momentum’s two roles cleanly (Table C.3, Appendix C.2): with the full buffer standing in for momentum’s averaging, C8 already has essentially no gap at momentum 0 (0.2 pp), so the residual C7 leaves is variance, closable by source variance reduction rather than by acceleration; adding momentum to C8 then lifts ACC to 0.955 while re-introducing only a 2.0 pp transient. The headline is that *the curriculum needs momentum for accuracy; its residual gap needs only variance reduction, and any source of it will do*. The full 60k buffer defeats the limited-memory premise and is diagnostic only; the practical configuration is C7_M (3.7 pp / 0.03 / 0.931) at 1/60th the memory. For completeness the cross also confirms the vanilla-ER null: momentum shifts vanilla depth only from 19.5 to 15.9 pp, smaller than the seed spread with seeds split on its sign ($p = 0.44$), because the deterministic unopposed first step carries no accumulated velocity to average; it does cut vanilla’s tail (area $6.3 \rightarrow 1.0$, all seeds). Momentum smooths vanilla’s recovery, but it leaves the depth, the gap itself, intact.

6.3 The Directional Feedforward Gate (RQ3)

RQ3 asks whether asymmetric PER shrinks the gap as its filter strengthens ($\delta \rightarrow 0$), at what plasticity cost, and where a feedforward gate fails. Table 6.2 reports the δ sweep on both buffer legs at momentum 0; the full metric set at both momentum settings is in Appendix C.4. The standard 1k leg is the gate’s prac-

Condition	Momentum 0.0			Momentum 0.9		
	gap_depth (pp)	gap_area_end	ACC	gap_depth (pp)	gap_area_end	ACC
Vanilla ER (baseline)	19.5 ± 4.1	6.3 ± 0.5	0.873 ± 0.005	15.9 ± 2.6	1.0 ± 0.2	0.928 ± 0.003
C1: Linear $N=50$	17.5 ± 1.7	4.9 ± 0.5	0.851 ± 0.023	6.4 ± 0.5	0.2 ± 0.1	0.932 ± 0.003
C2: Linear $N=100$	16.2 ± 4.3	3.6 ± 0.3	0.844 ± 0.025	6.4 ± 0.6	0.2 ± 0.1	0.932 ± 0.003
C3: Linear $N=200$	14.3 ± 4.6	1.8 ± 0.6	0.828 ± 0.025	6.2 ± 0.6	0.1 ± 0.0	0.925 ± 0.007
C4: Adaptive	12.8 ± 2.9	2.2 ± 0.5	0.833 ± 0.025	2.5 ± 0.5	0.3 ± 0.1	0.917 ± 0.001
C5: Adaptive $\lambda_{\min}=0.05$	13.2 ± 3.2	2.2 ± 0.5	0.835 ± 0.025	2.6 ± 0.5	0.1 ± 0.0	0.922 ± 0.001
C6: Adaptive $\lambda_{\min}=0.10$	13.9 ± 3.3	2.4 ± 0.5	0.836 ± 0.026	3.0 ± 0.4	0.0 ± 0.0	0.927 ± 0.002
C7: Adaptive $\lambda_{\min}=0.20$	13.9 ± 3.4	2.7 ± 0.6	0.837 ± 0.031	3.7 ± 0.4	0.0 ± 0.0	0.931 ± 0.002

Table 6.1: λ -curriculum sweep on rot-MNIST against vanilla ER, at momentum 0.0 and 0.9; five seeds. The scalar feedback gate reaches a few points of depth at matched accuracy only in the momentum-on regime it is intended for.

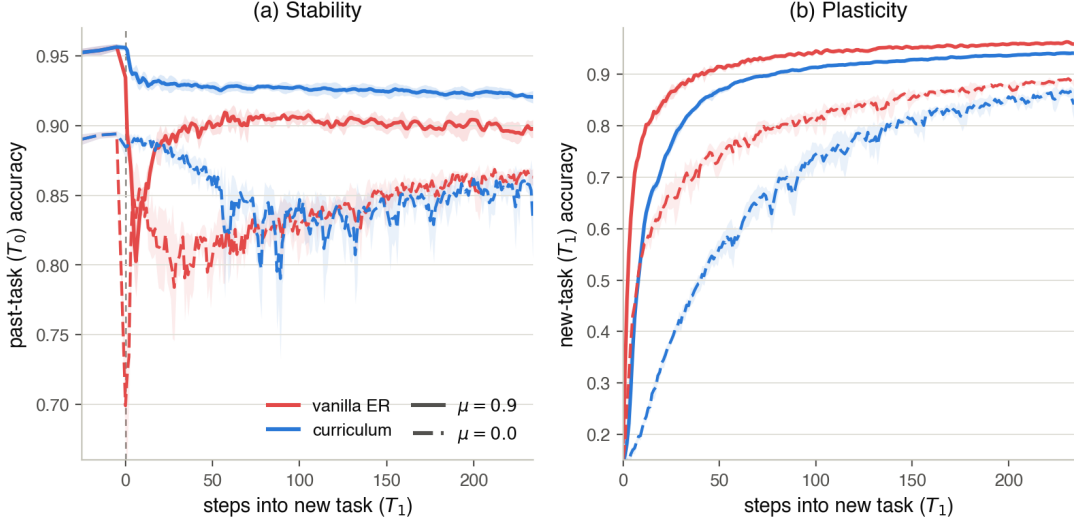


Figure 6.2: The scalar feedback gate and momentum’s two roles: per-step accuracy across the rot-MNIST switch, seed mean with $\pm$ std bands, for vanilla ER (red) and the practical adaptive curriculum $\lambda_{\min}=0.20$ (blue), each at momentum 0.9 (solid) and 0.0 (dashed). (a) Stability (T_0): only curriculum-with-momentum (solid blue) holds the past task near its plateau through the switch; momentum barely dents vanilla’s spike (solid red still dives), and both methods open a deep, jittery gap without it (dashed). Momentum closes the gap where the gate has already tempered the push at source, not otherwise. (b) Plasticity (T_1): at momentum 0 the curriculum slows early new-task learning (dashed blue is slowest to rise, the plasticity debt of holding the push down), but momentum pays it back, lifting the solid-blue curve back toward the vanilla pace.

tical configuration; the full-buffer leg plays the same diagnostic role as C8, making the replay gradient exact so that the remaining area is the arc, and it does not enter the head-to-head comparisons.

Monotone, and at no plasticity cost. On the standard buffer, depth falls from 9.0 to 4.8 pp as δ goes from 1.0 to 0.03 while ACC rises from 0.869 to 0.880 and WP_{100} from 0.733 to 0.741; every one of these trends holds across all five seeds (Figure 6.3(a)). The roughly 4 pp of accuracy the scalar gate pays at momentum 0 never materialises here, because the filter blocks the push only in the directions the past task is steep in and lets it through everywhere else. This is the scalar gate’s weakness inverted, and it is what makes the two gates complements rather than a strong and a weak version of the same thing. The full-

buffer leg isolates the arc: with the variance driver off, the area shrinks with the filter, $2.3 \rightarrow 1.0 \rightarrow 0.4$ for $\delta = 1.0 \rightarrow 0.3 \rightarrow 0.1$ (the $1.0 \rightarrow 0.3$ drop holds across all seeds), the independent level-1 evidence promised in Section 6.1: the directional gate bends the followed path back toward the reference, as the curriculum does by another route. On the same diagnostic leg the strongest cliff-free filter ($\delta=0.3$) takes depth to 1.5 pp, under even the driver-removal floor (D4: 1.9 pp) — a mechanism reading, not an operating point: it says the gate reaches the arc itself, which no driver removal does, while the full buffer that isolates it defeats the limited-memory premise. Leaving the restoring gradient g_{rep} at full strength is what makes this a net removal of damage rather than a reshuffling of it: a symmetric variant that preconditions the summed update strangles the restoring gradient harder than

δ	Standard (1k) buffer				Full (60k) buffer			
	depth (pp)	area	ACC	WP ₁₀₀	depth (pp)	area	ACC	min-ACC
1.0	9.0 ± 0.6	3.3	0.869	0.733	4.0 ± 1.0	2.3	0.898	0.841
0.3	7.4 ± 1.1	2.3	0.873	0.736	1.5 ± 1.1	1.0	0.900	0.867
0.1	5.9 ± 1.5	1.6	0.876	0.739	2.2 ± 1.6	0.4	0.901	0.860
0.03	4.8 ± 1.9	1.0	0.880	0.741	9.5 ± 8.2	0.4	0.903	0.787

Table 6.2: Asymmetric-PER δ sweep on rot-MNIST at momentum 0, five seeds. Depth and area fall as $\delta \rightarrow 0$ while ACC and WP₁₀₀ rise: the directional gate protects the past task at no plasticity cost. On the full buffer, a diagnostic leg where the area is the deterministic arc, the arc shrinks with the filter, the strongest cliff-free setting ($\delta=0.3$) undercutting even the driver-removal floor (D4: 1.9 pp). The far cell is the exception: at $\delta=0.03$ full buffer the mean depth jumps to 9.5 pp with a large std, the cliffs of the text.

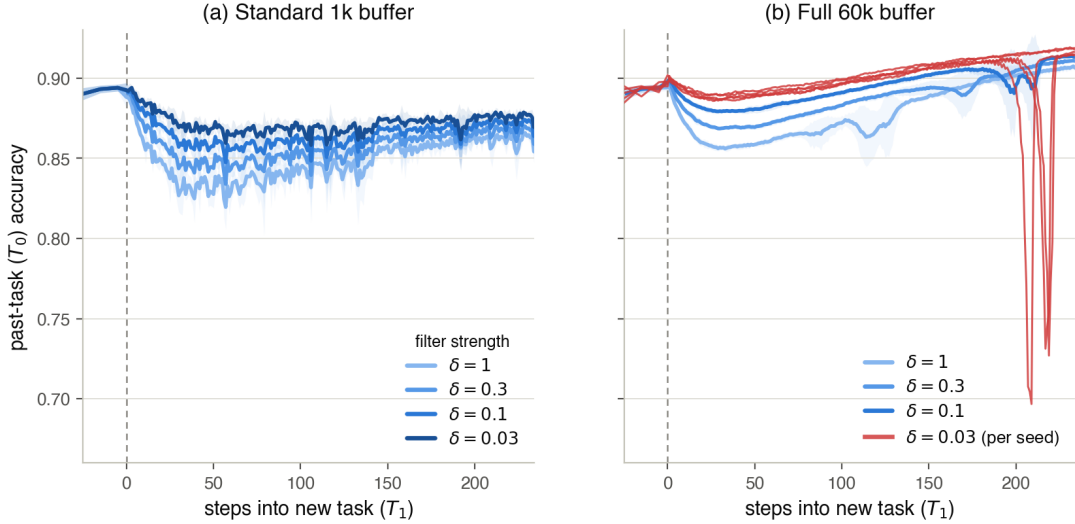


Figure 6.3: The directional feedforward gate: per-step past-task (T_0) accuracy across the rot-MNIST switch at momentum 0, sweeping the damping δ (weaker filter light $\rightarrow$ stronger filter dark). (a) Standard 1k buffer: the boundary dip shrinks monotonically as $\delta \rightarrow 0$, and accuracy rises with it (Table 6.2), so the protection is bought at no plasticity cost. (b) Full 60k buffer (variance driver off, so the residual excursion is the deterministic arc): the arc shrinks with the filter for $\delta \geq 0.1$, but at the strongest setting $\delta=0.03$ the gate fails late and abruptly: shown per seed (critical red), 3 of 5 seeds hold flat for ~ 200 steps and then slide off a late cliff, a failure mode a feedback signal does not share.

the harmful push, so its forgetting *climbs* as the filter strengthens (Appendix C.5, Table C.6).

At momentum 0 the directional gate dominates the scalar gate. Head to head at momentum 0, the strongest standard-buffer setting ($\delta=0.03$: 4.8 pp / 0.880) beats the practical curriculum (C7: 13.9 pp / 0.837) on both axes at once, lower in depth and higher in accuracy. Part of this is that the curvature filter also limits how far any single noisy gradient can move the past-task loss, so it suppresses the per-step *expression* of buffer variance, which the scalar gate can only counter with momentum or a larger buffer. The scalar gate’s case is not this corner; it is momentum composability, which the next results show the directional gate lacking, and robustness to the failure mode reported next.

The cliffs: an unexpected late failure. The one place the sweep breaks monotonicity is the strongest

full-buffer setting, $\delta = 0.03$, where mean depth jumps to 9.5 pp with a std of 8.2: in 3 of 5 seeds the past-task curve holds flat for roughly two hundred steps and then slides off a late cliff, the red per-seed traces in Figure 6.3(b). A solver control rules out the most straightforward explanation: rerunning the setting with the conjugate-gradient solver at 60 iterations instead of 25 reproduces the cliffs at the same seeds and steps (11.5 pp mean depth), so they are not an artefact of an under-converged solve. The cliffs came as a surprise: the account made no prediction of a late, abrupt failure, and nothing in the rest of the sweep hints at one. They are also confined to the exact-gradient diagnostic: at the same δ on the standard 1k buffer no seed shows one (depth 4.8 ± 1.9 pp, the monotone end of the sweep). The failure is tied to the exact replay gradient: the sampling jitter of a practical buffer appears to protect against it, the same noise that elsewhere drives the overshoot here keeping the iterate from settling into the escape direction.

Why the cliffs happen we can only partly say. A reading consistent with the data is a blind spot of the gate’s model of the past task: the filter passes new-task pressure through directions its sampled Fisher rates as flat, and in these seeds that trust appears to be misplaced further out along the trajectory. We offer this as a hypothesis, not an established mechanism. What the control does establish is that the strongest filter can fail late and abruptly when nothing perturbs it. Because the failure is confined to the diagnostic leg, it is unlikely in a practical configuration, but it exposes a real weakness of feedforward control.

Momentum re-inflates the per-step filter. Momentum undoes the directional gate’s protection, the opposite of what it does for the scalar gate, and it does so on both legs: on the practical 1k buffer depth climbs from 4.8 to 8.5 pp at $\delta=0.03$ (Figure 6.4), and on the full-buffer diagnostic from 2.2 to 7.6 pp at $\delta=0.1$ and from 1.5 to 9.1 pp at $\delta=0.3$ (both across all five seeds). Momentum accumulates the already-filtered direction, but the filter attenuates the step, not the source: the new-task loss, and with it the per-step gradient, keeps its full size, and because motion in the braked directions is slow, the shrunk push persists there step after step. A persistent push entering the accumulator is amplified toward $1/(1-\mu)$ times its per-step size, so velocity rebuilds displacement in the very directions the filter shrinks one step at a time. At the practical buffer this leaves asymmetric PER under momentum at vanilla-level accuracy (0.929) with a depth (8.5 pp) between the scalar gate’s (3.7 pp, $C7_M$) and vanilla ER’s (15.9 pp). This re-inflation, not a fixed ordering, is what makes the head-to-head comparison of the next subsection regime-dependent.

6.4 Reading the Momentum Crosses Together

This subsection answers no separate research question; it reads the momentum crosses of RQ2d and RQ3 side by side, in both gates’ practical regime, the 1k reservoir, with the full-buffer legs staying in their diagnostic role. Together the crosses show that the two gates do not sit on a fixed stability–plasticity frontier that a practitioner trades along: which gate leaves the smaller gap is decided by the momentum setting, the same setting that decides final accuracy. Figure 6.4 shows it directly: the same three methods at the same buffer, with only the momentum setting changed between the panels.

No momentum: the directional gate is better on both axes. At momentum 0 and a matched 1k buffer, the directional gate beats the scalar gate on both axes (Figure 6.4(a)). Its strongest standard-buffer setting reaches 4.8 pp of depth at 0.880 accu-

racy (Table 6.2) against the curriculum’s 13.9 pp at 0.837 ($C7$, Table 6.1): lower gap *and* higher accuracy, because the per-step filter protects the past task without ever slowing the new one at source, so it carries no plasticity debt for the momentum-0 regime to leave unpaid. The one cost that remains is compute: the protection is bought with a conjugate-gradient solve in the replay Fisher at every step, against the scalar gate’s single gradient-norm ratio (Table C.10). When momentum is off and the per-step solve is affordable, the directional gate is the better choice.

Momentum: the scalar gate takes over. Momentum is required for accuracy on any realistic backbone, and it reverses the ranking through the re-inflation mechanism of Section 6.3: the directional gate’s depth climbs from 4.8 to 8.5 pp at the matched buffer (Figure 6.4(b)), while the scalar gate composes with momentum because it attenuated at source, and reaches 3.7 pp at vanilla accuracy ($C7_M$). Once the buffers match, the directional gate has no accuracy advantage left to offset its re-inflated gap (0.929 vs. the curriculum’s 0.931); the scalar gate is simply the lower-gap method in the practically relevant regime. The choice between the gates is therefore not a point on a frontier but a consequence of whether momentum is on: off, take the directional gate; on, take the scalar gate. A five-task rot-MNIST sequence reproduces this picture across four consecutive boundaries: at momentum 0 both gates hold the first task nearly flat through every switch, with the directional gate keeping its edge (the highest final accuracy, 0.870, and the smallest per-boundary area); under momentum the curriculum keeps the least cumulative forgetting while the re-inflated directional gate tracks vanilla (Appendix C.9, Figure C.2).

6.5 Generalisation to CIFAR-10 (RQ4)

The CIFAR-10 block tests whether both gates carry to a ResNet-18, a clean $\rightarrow$ corruption shift, and the two normalisations that govern the post-switch recovery. Table 6.3 reports the headline block at momentum 0.9, the practically relevant regime.

Both gates carry to the deeper backbone. The rot-MNIST result carries over and strengthens. Under batch norm the curriculum cuts depth from 11.8 to 4.8 pp and asymmetric PER to 5.9 pp; under group norm, where vanilla ER suffers a catastrophic 41.2 pp drop, the curriculum cuts it to 5.2 pp and asymmetric PER to 5.6 pp. Every gate-vs-vanilla depth and area contrast holds across all five seeds, at final accuracy within about 0.01 of vanilla and with worst-case retention sharply improved, min-ACC rising from 0.31 to 0.67 under group norm. On this backbone the transient stays open for hundreds of steps, so the area

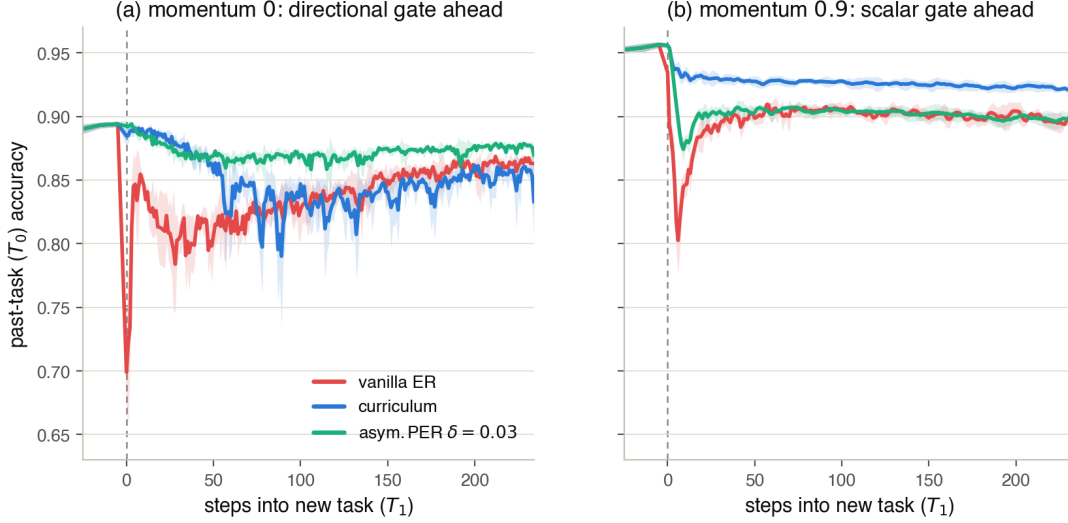


Figure 6.4: The momentum synthesis at the matched 1k buffer: per-step past-task (T_0) accuracy across the rot-MNIST switch for vanilla ER (red), the practical curriculum C7 (blue), and asymmetric PER at its strongest standard-buffer filter $\delta=0.03$ (aqua); seed mean with $\pm$ std bands, $x=0$ the task switch. (a) Momentum 0: the directional gate holds T_0 highest (4.8 pp / ACC 0.880) while the curriculum opens a deep, noisy dip (13.9 pp / 0.837). (b) Momentum 0.9: the ordering swaps: the curriculum holds T_0 near its plateau (3.7 pp / 0.931) while accumulated velocity re-inflates the directional gate’s per-step filter (8.5 pp / 0.929). The pre-switch level is higher in (b) because momentum converges T_0 higher before the switch.

Norm	Method	gap_depth (pp)	gap_area_end	ACC	min-ACC
BatchNorm	G1: Vanilla ER	11.8 ± 2.5	17.5 ± 4.6	0.723 ± 0.008	0.644 ± 0.023
	G2: Curriculum ($\lambda_{\min}=0.20$)	4.8 ± 0.7	3.6 ± 2.2	0.722 ± 0.011	0.713 ± 0.017
	G3: Asym. PER ($\delta=0.1$)	5.9 ± 4.3	3.3 ± 3.0	0.715 ± 0.011	0.698 ± 0.023
GroupNorm	G4: Vanilla ER	41.2 ± 13.5	34.9 ± 23.7	0.730 ± 0.011	0.309 ± 0.111
	G5: Curriculum ($\lambda_{\min}=0.20$)	5.2 ± 2.1	1.7 ± 0.9	0.731 ± 0.020	0.670 ± 0.055
	G6: Asym. PER ($\delta=0.1$)	5.6 ± 1.1	4.8 ± 3.8	0.741 ± 0.009	0.673 ± 0.023

Table 6.3: CIFAR-10 headline block (ResNet-18, momentum 0.9, buffer 5,000, five seeds). Both gates cut gap depth and area far below vanilla ER at matched final accuracy and much higher worst-case retention (min-ACC), under both normalisations. Neither was given a CIFAR-specific sweep; both carry their rot-MNIST configurations.

carries as much of the comparison as the depth (Figure 6.5): vanilla ER under group norm crashes T_0 almost to chance and crawls back over the whole recovery window, while both gates hold T_0 near its plateau through the switch. At the matched 5,000 buffer and momentum 0.9 both gates land in the same few-pp band, with the curriculum the cheaper of the two (below), consistent with the momentum regime of Section 6.4.

Mechanism, generalisation, and cost. Three further checks are reported in full in Appendix C.8. A momentum cross on the deep backbone (Table C.9) reproduces two of the three predicted signatures, momentum leaving vanilla depth unchanged ($40.5 \rightarrow 41.2$ pp) and composing with the curriculum ($28.6 \rightarrow 5.2$ pp), while the directional-gate re-inflation is confounded by momentum-0 under-convergence of the ResNet (vanilla min-ACC 0.11), so the re-inflation signature rests on the rot-MNIST result where the

full-buffer MLP isolates it cleanly. A generalisation block (Appendix C.7, Table C.8) shows the curriculum beating a paired vanilla control on two corruptions it was never tuned on (shot noise $46.3 \rightarrow 7.3$ pp, contrast $8.4 \rightarrow 3.9$ pp under group norm), with a single three-task group-norm sequence flagged as a reliability boundary of the adaptive ratio when two consecutive corruptions are near-identical and both gradient norms become small (Figure C.3; traced in Appendix C.7). Finally, compute separates the gates where retention does not (Table C.10): the scalar gate is one gradient-norm ratio on top of a standard ER step and leaves the GPU half-idle, while the directional gate’s 25 conjugate-gradient Fisher-vector products per step saturate it, costing $\sim 16\times$ the wall-clock and $\sim 24\times$ the GPU energy on the ResNet-18 at similar peak memory; on the MLP both runs finish in under half a minute and the difference is negligible. In the momentum regime that practice requires, then, the scalar gate matches the directional gate’s

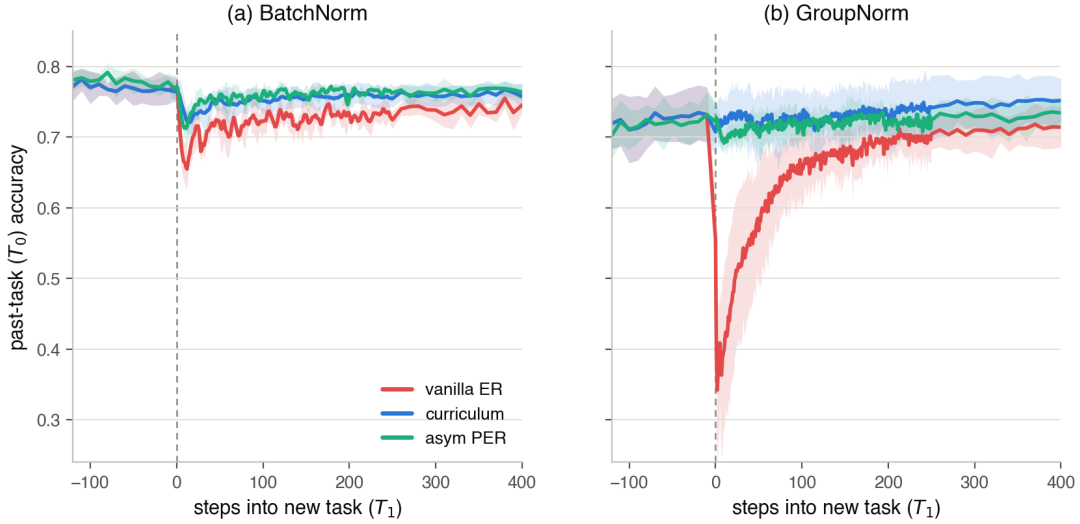


Figure 6.5: Per-step past-task (T_0) accuracy across the CIFAR-10 transition (ResNet-18, momentum 0.9, buffer 5,000), one panel per normalisation, each overlaying vanilla ER (red), the practical curriculum (blue) and asymmetric PER (aqua); seed mean with $\pm$ std bands, $x=0$ the task switch. (a) BatchNorm: vanilla ER dips and re-adapts slowly while both gates hold T_0 near its plateau. (b) GroupNorm: vanilla ER’s transient is deep and long, whereas both gates hold T_0 flat through the switch (Table 6.3, Appendix C.6).

retention at essentially vanilla-ER cost and without the blind-spot cliffs, which is why it is the default recommendation.

6.6 Summary of Findings

The account holds end to end. (i) The gap separates into a deterministic arc and overshoot of the followed path: removing the drivers leaves the arc, fixing the path leaves the overshoot, doing both closes the gap, and the permanent stability–plasticity cost remains. (ii) The scalar feedback gate, with momentum, reaches 3.7 pp of depth at vanilla accuracy: momentum repays the plasticity the schedule borrows, and its averaging supplies the variance reduction the residual gap needs. (iii) The directional feedforward gate shrinks depth and arc monotonically as $\delta \rightarrow 0$ at no plasticity cost, but fails late and abruptly at its strongest diagnostic setting (ruled out as a solver artefact, read as curvature blind spots) and re-inflates under momentum. (iv) The comparison is therefore regime-dependent, not a frontier: without momentum the directional gate is better on both axes; with momentum, required for accuracy, the scalar gate takes over at vanilla-ER cost. Both gates carry to a ResNet-18 on CIFAR-10 under both normalisations at matched accuracy and much improved worst-case retention. The discussion reads these findings against the account of Section 3.

7 Discussion

The results support the two-level account and, with it, the gate view of replay. This section reads them back against the theory: what the two levels explain that a single additive decomposition could not (Section 7.1), how the two gates divide the stability–plasticity work along the feedback/feedforward and scalar/directional axes (Section 7.2), what the account changes in the literature it grew out of (Section 7.3), the limits of the evidence (Section 7.4), and where the taxonomy points next (Section 7.5).

7.1 Why the gap needs two levels

The central empirical claim is that the stability gap is not one quantity but two failures of the same optimisation, kept apart because they answer to different interventions. The 2×2 of Section 6.1 is the direct test. Removing the overshoot drivers by construction leaves a deterministic arc (D4: 1.9 pp of depth but a clear bend, area 3.2, with the replay gradient exact and the buffer noiseless); fixing the path with the curriculum leaves the overshoot (C7 at momentum 0: 13.9 pp, essentially all of it); only doing both closes the gap (C8: 0.2 pp). Neither intervention is a weaker version of the other. The arc is a property of which path the descent dynamics define, and no amount of optimiser hygiene, exact gradients, balanced magnitudes, zero noise, touches it; the overshoot is how far the discrete, noisy, accelerated iterate leaves whatever path it is on, and no change of path removes it. This is why the levels must be named separately: an inter-

vention aimed at one is silent on the other.

The natural alternative, a single additive decomposition of the gap into a magnitude share, a variance share, and a trajectory share, does not survive the data. The shares are order-dependent (a noisy replay gradient inflates the first-step magnitude, so the factors are not separable summands), and the mapping from factors to metrics is crossed by the experiments: the curriculum’s momentum-0 tail looks like a “trajectory” term until C8 removes it with the exact buffer and no momentum at all (Appendix C.2, Table C.3), revealing it as variance. The two-level split is by kind, which path is followed versus how tightly it is followed, and that is the distinction the interventions respect.

The split also settles what the gap is not the price of: C8 closes the transient to 0.2 pp while its accuracy sits 5.5 pp below vanilla ER (Section 6.1), so the permanent rise to the joint-basin level is the stability–plasticity cost and everything below the settled level and back is failure to follow the monotone reference path, not the cost of learning. This is why the end-referenced area is anchored to the recovered plateau rather than the pre-switch baseline: it attributes the gap to the trajectory rather than to capacity loss.

Finally, the two levels explain why the metric that matters changes with the backbone. On the MLP the arc is cheap: the network reconverges in a few steps, so the open bend integrates to little and the perceived severity is the overshoot spike. On the ResNet the same transient stays open for hundreds of steps (Figure 6.5), so the slow tail, the part that persists, is where most of the accuracy the gap costs accumulates, and it is where both gates’ margin over vanilla is largest (Table 6.3). Depth and area are not two views of one number; they are the signatures of the two levels, and which one dominates the cost is set by how slowly the network recovers. A depth-only reading would have understated the result on exactly the architectures where continual-evaluation cost is highest.

7.2 Two gates, two ways to divide the work

Written in the unified update of Section 3.3, every replay method gates the new-task push while restoring at full strength, and the curriculum and asymmetric PER are the two pure corners of the taxonomy: a scalar gate driven by feedback, and a directional gate driven by feedforward. The results show the two corners working and failing exactly where the taxonomy says they should, and the single fact that organises the whole comparison is *where* the gate attenuates the push.

The scalar feedback gate attenuates at source. It multiplies the entire new-task loss by a schedule that starts near zero, so early in the transition there is al-

most no push to discretise, accumulate, or let noise act on, and by the time the weight has grown the trajectory has already settled toward the joint basin. This has three consequences the data confirm. It composes with momentum: because the push is small and clean at source, momentum finds a stochastic residual to average and closes the gap it left untouched under vanilla ER (Section 6.2), while the accumulator has no sustained full-strength direction to rebuild. It has no curvature blind spot: the adaptive schedule reads the live replay-gradient norm, a signal that fires whichever direction the damage arrives from, so it cannot be steered past a threat it did not model. And it pays a plasticity debt: holding the loss down slows the new task, which is why C8 at momentum 0 is gap-free but 5.5 pp down on accuracy, and why the debt is repaid only by acceleration, momentum lifting C8 from 0.818 to 0.955 (Table C.3). The scalar gate’s price is plasticity, and momentum is how it is paid.

The directional feedforward gate attenuates per step. It filters the push through the replay Fisher every step, passing motion where the past task is flat and braking it where the past task is curved, and it never slows the new task at source, so it protects the past task at no plasticity cost, ACC and short-horizon plasticity rising as the filter strengthens (Table 6.2). The same per-step locality is its weakness on two axes, one the taxonomy predicts and one that emerged unanticipated. Under momentum the protection unravels: the driving magnitude of the new-task gradient is untouched, so velocity accumulated across steps reconstructs displacement in the very directions the filter shrinks one step at a time, and depth re-inflates from ≈ 2 to ≈ 8 pp (Section 6.3). This is the same lever as the curriculum, momentum, producing the opposite outcome, and the difference is entirely one of where the attenuation was applied: source versus step. That single distinction accounts for the whole momentum column of the study, the vanilla depth null (no accumulated velocity in the deterministic first step), the curriculum’s composition (source attenuation), and the directional gate’s re-inflation (step attenuation).

The directional gate’s second failure was not predicted; it surprised us, and it is the sharper one because it makes the case for feedback. At the strongest filter the past-task curve can slide off a late cliff (the $\delta=0.03$ full-buffer cliffs, reproduced under a stronger solver and absent on the practical buffer, whose gradient jitter appears to protect against them; Section 6.3). A plausible reading is a blind spot of its curvature model: the sampled Fisher rates a direction flat that the true past-task loss punishes further out, and new-task pressure escapes through it. In hindsight this exposure is characteristic of feedforward control: a feedforward gate can only be as good as its model of the past task, and that model is estimated on finite data under a saturating loss. A feedback gate has no

such blind spot because it does not model the damage in advance, it reads it; its own weakness lies elsewhere, in the conditioning of the signal it reads, as the three-task sequence shows (Section 7.4). The cliffs and the plasticity debt are therefore the two gates’ defining costs, and they are complementary: the feedforward gate is cheap in plasticity but myopic, the feedback gate is harder to blindside but slow. This is why the two do not sit on a fixed frontier but flip with momentum (Section 6.4). At a matched memory budget and no momentum the directional gate wins on both axes, because it pays no plasticity and momentum has not yet unwound its filter. Momentum, required for accuracy, then re-inflates the directional gate’s per-step filter while composing with the scalar gate at source, so the scalar gate becomes the lower-gap method in the practically relevant regime, and does so at vanilla-ER compute against the directional gate’s per-step Fisher solve (Section 6.4). Both gates carry to the ResNet at matched accuracy and much improved worst-case retention (Table 6.3), and the two mechanism signatures not confounded by convergence reproduce there (Appendix C.8, Table C.9). It is therefore the gate principle, not either particular method, that generalises.

7.3 What the account changes in the literature

The two-level split settles what reads as a disagreement between the two accounts of the gap that Section 2.2 set out. Lange et al. [16] trace the drop to the gradient imbalance at the boundary; Hess et al. [11] show the gap survives training on the exact joint loss, and Kao et al. [14] derive the geometry that makes it do so. Each account is right about one level and silent on the other. The imbalance is the acute driver of the overshoot: balancing it away is by far the largest single lever on depth in the driver ladder ($19.5 \rightarrow 5.7$ pp), and it leaves the arc untouched. The persistence under exact gradients is the arc: D4 reproduces it under exact, balanced, noiseless replay and measures what remains, 1.9 pp of depth and an area of 3.2, a controlled counterpart of the joint-loss observation and a quantification of the analytical mechanism. Read this way the two accounts stop competing. Most of the measured depth is the overshoot the imbalance account describes; most of what no repair of the update can remove is the geometry the trajectory account describes; and a mitigation aimed at either alone leaves the other level standing, which is the practical content of the 2×2 .

The unified update of Sodhani et al. [22] sorts replay methods by which weight they move and from what signal; the results add the axis that turned out to be decisive, where the modulation is applied. A-GEM and MEGA hold the new-task weight at one and correct per step from a live signal [2, 8], which

places them on the step side of the source/step divide, the side whose protection momentum unwinds in Section 6.3. The account therefore predicts that their boundary protection weakens under momentum as the directional gate’s does, while source-attenuating schedules compose with it. These results do not test that prediction, but it is directly testable, and it matters because the original comparisons predate the continual-evaluation lens: end-of-task metrics would not have registered per-step protection eroding, any more than they register the gap itself.

Mode connectivity enters the account as an existence result and exits as a control result. Garipov et al. [6] and, for the continual setting, Mirzadeh et al. [20] establish post hoc that the relevant solutions share a low-loss corridor; the corridor is inspected after training and says nothing about whether an optimiser can be kept inside it while learning. C8 supplies the dynamic counterpart: an update rule whose iterate tracks the monotone reference path through the transition (0.2 pp of depth), turning the observation that a benign route exists into the statement that a gate can make training follow it. The same experiment sharpens the stability–plasticity dilemma [19] into a measurement. At fixed capacity and memory, the dilemma’s price is the permanent 5.5 pp offset to the joint level; the transient that continual evaluation reveals, up to 19.5 pp at the boundary here, is not part of that price but a routing failure the right gate removes. The dilemma bounds where the model can settle; it does not oblige the model to swing below that level on the way.

7.4 Limitations

Several limits bound these claims. The first is statistical: five seeds floor the two-sided Wilcoxon test at $p = 0.0625$, so the argument rests on effect sizes and seed-paired direction, not a 0.05 threshold the design cannot reach. The effects the story leans on, the up-to-eightfold depth reductions on CIFAR, the C7→C8 variance collapse, the momentum re-inflation of the directional gate, are large relative to the seed spread and unanimous across seeds; the smaller ones, the adjacent- δ steps and the $\lambda_{\min}$ trade-offs, are read as directional consistency only, and one, the adaptive-versus-best-linear depth, does not reach even that ($p = 0.44$).

The second is the accuracy drift of the scalar gate at momentum 0: lengthening the linear ramp lowers depth but costs several points of final accuracy (Table 6.1), so the method’s accuracy-neutrality is a property of the full curriculum-plus-momentum configuration, not of the ramp alone. The third is the directional gate’s cliffs: the solver control establishes that they are real and not a numerical artefact, but the blind-spot mechanism we offer for them is a reading consistent with the data, not an established

cause. The fourth is the CIFAR momentum cross: the directional-gate re-inflation signature does not reproduce on the ResNet, and while we attribute this to momentum-0 under-convergence of the backbone rather than to a failure of the mechanism (which the MLP isolates cleanly), a clean deep-backbone test of that one signature would require a converged momentum-0 baseline we did not run. The fifth is the three-task group-norm sequence, where the adaptive ratio loses conditioning at a boundary between two near-identical corruptions and the schedule underperforms its control with high seed variance; we flag it as a reliability boundary of the gradient-ratio feedback signal, absent from the batch-norm variant, rather than a property of the gate principle; Appendix C.7 traces the failure step by step through the logged gradient-norm ratio.

Two scope limits are worth stating plainly. The full-buffer legs of both the driver block and the δ sweep store the entire past task and so break the limited-memory premise of continual learning; they are diagnostic probes that isolate the arc and split momentum’s roles, and the practical configurations are the 1k-buffer curriculum (C7_M) and the standard-buffer directional gate. And the directional gate’s per-step Fisher-vector solve adds a fixed multiple of the backward cost per step (25 conjugate-gradient iterations here), a real overhead on large backbones, $\sim 16\times$ the wall-clock and $\sim 24\times$ the GPU energy of the curriculum on the ResNet-18 (Table C.10), that the scalar gate, a single changing scalar with no stored curvature, does not carry: it adds one gradient-norm ratio to a standard ER step. The headline analysis also rests on a single-transition, two-task design chosen to isolate one boundary; the five-task rot-MNIST sequence (Appendix C.9) shows the regime picture surviving four consecutive boundaries, but a systematic study of longer sequences and of class-incremental rather than domain-incremental shifts is left open.

7.5 Outlook

The taxonomy that organises the two methods also names what is missing from it. The gates studied here fill three of its four corners, scalar $\times$ feedback, directional $\times$ feedforward, and, as an ablation, the linear ramp at scalar $\times$ feedforward. The remaining corner, a directional gate driven by a live damage signal rather than by a precomputed curvature model, is exactly the design the cliffs argue for: it would filter the push by direction, keeping the plasticity that makes the feedforward gate cheap, while reading the damage as it arrives, closing the blind spot that makes it myopic. The momentum results constrain its design in advance: applied to the instantaneous step it would inherit the re-inflation of Section 6.3, so the hybrid must place its directional attenuation at source or on the accumulated velocity. Building and testing such a

gate is the most direct consequence of these results.

Three narrower directions follow from the failure modes and costs. The directional gate’s momentum re-inflation is a per-step-attenuation artefact; a version that attenuates the accumulated velocity rather than the instantaneous step, or that damps the push at source in the directions the Fisher flags, would be predicted to compose with momentum the way the scalar gate does, a testable claim. The stakes of that claim reach past this one method: any approach that preconditions the update with past-task curvature per step, the natural-gradient family included [14], leaves the driving magnitude of the new-task gradient intact and so carries the same exposure, and on backbones where momentum is not optional, moving the attenuation to the velocity is the general repair to test. The gate’s compute cost is likewise not fundamental: the per-step conjugate-gradient solve could be replaced by a cheaper Fisher approximation, diagonal or Kronecker-factored, and whether the protection survives the coarser metric is an open empirical question; on backbones much larger than the ResNet-18 here, where a fixed multiple of the backward cost per step is prohibitive, such an approximation is a precondition for the directional corner rather than a refinement of it. And momentum’s gap-closing role was shown to be interchangeable with source variance reduction (C8), which suggests the curriculum could be paired with other variance reducers: iterate averaging, which carries none of momentum’s accumulator-driven overshoot, could close the stochastic residual without the small transient price acceleration re-introduces. On the practical side, the regime dependence gives a clean rule. With momentum off the directional gate is better on both stability and accuracy, but momentum is needed for accuracy on any realistic backbone, and there the scalar gate closes the gap at almost no final-accuracy penalty, negligible overhead, and no blind-spot cliffs, while the directional gate re-inflates and carries a per-step Fisher cost. The scalar gate is therefore the default recommendation for systems trained with momentum and queried during training; the directional gate is reserved for the momentum-free regime where it dominates. The cliffs add a methodological rule to that practical one. They arrive late and abruptly, after long stretches of exemplary retention, and end-of-task metrics would have recorded the $\delta=0.03$ runs as successes; only per-step measurement caught the failure. Continual evaluation [16] is usually presented as the lens that reveals the stability gap; these results argue for it also as an acceptance test, run before any feedforward gate is trusted in a system that serves predictions while it trains, because a gate that acts on a model of the past task can fail precisely where that model is wrong, silently until it does.

Statement on the Use of AI Tools

In accordance with the University of Groningen’s guidelines on the responsible use of generative AI, I disclose that AI assistants were used during the preparation of this thesis. The tools were Google Gemini Pro (version 3 Pro, then version 3.1 Pro from February 2026) and Anthropic Claude Opus (the 4.x line, versions 4.5 through 4.8) — the releases current across the roughly six-month period, January to July 2026, in which the work was carried out. They functioned as assistive tools under my continuous direction, not as autonomous contributors. Specifically, I used them to:

- brainstorm and stress-test ideas, and as a sounding board to reason through arguments and explanations;
- draft and revise prose, which I subsequently edited for accuracy, precision, and voice;
- assist in writing and refactoring the analysis and experiment code, all of which I reviewed, ran, and tested;
- produce first-pass summaries of the literature, which I verified against the primary sources.

The tools were never given open-ended control: every request specified a concrete, bounded task, and all output was reviewed and, where retained, corrected, understood, and integrated by me. All experimental results, figures, and numerical claims were checked against the underlying run data, and all cited work was read in the original. The research questions, experimental design, interpretation of the results, and the overall argument of the thesis are my own, and I take full responsibility for its content, including any errors.

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A Implementation Details

This appendix records the implementation specifics that the methods of Section 5 depend on but that would interrupt the argument in the main text: the datasets, the backbones and training protocol, the experience-replay variants, and the asymmetric-PER solver.

A.1 Datasets

rot-MNIST rotates each 28×28 image by the per-task angle and feeds the flattened greyscale vector to the MLP. The CIFAR-10 domain shifts are generated with the `imagecorruptions` package, which implements the standard common-corruptions benchmark of Hendrycks and Dietterich [10]; the Gaussian noise, shot noise, and contrast shifts are applied at severity 3 on the benchmark’s 1–5 scale, per sample at load time, followed by the standard CIFAR-10 channel normalisation. The clean task (“none”) applies the normalisation only.

A.2 Models and Training

Tables A.1 and A.2 give the two backbones, and Table A.3 the training protocol shared across both benchmarks.

Component	Value
Input	784 (flattened 28×28 greyscale)
Hidden layers	2×400 (fully connected, ReLU)
Output	10-class logits
Regularisation	none
Total parameters	478,410

Table A.1: MLP backbone used for the rot-MNIST sweeps.

Component	Value
Input	$3 \times 32 \times 32$ RGB
Stem	3×3 conv, stride 1, no max-pool (CIFAR adaptation)
Stages	4 residual stages, BasicBlocks [2, 2, 2, 2], widths [64, 128, 256, 512]
Normalisation	BatchNorm (default); GroupNorm (min(32, C) groups) variant crossed in Section 5.2.5
Head	global average pool + linear, 10-class logits
Total parameters	$\approx 11.2\text{M}$

Table A.2: CIFAR-adapted ResNet-18 backbone used for the CIFAR-10 block. The 3×3 stride-1 stem and removed max-pool keep the feature map at 32×32 ; all other details follow the ResNet-18.

Hyperparameter	rot-MNIST (MLP)	CIFAR-10 (ResNet-18)
Epochs per task	1 (online)	10
Batch size	256	256
Optimiser	SGD	SGD
Learning rate	0.1	0.1
Momentum	0.0 or 0.9	0.9 (0.0 in cross)
Weight decay	0.0	0.0
Replay buffer	1,000 or 60,000	5,000
Seeds	5	5 or 3

Table A.3: Training protocol for both benchmarks.

A.3 Experience replay

The replay buffer uses reservoir sampling [26], so the stored set is a uniform sample of the past stream at any budget. Three modes appear in the experiments:

- *Standard ER* (the D1 and all C-series conditions): each step takes one combined backward pass on $L_{\text{current}} + L_{\text{replay}}$, with half the mini-batch drawn from the current task and half sampled with replacement from the buffer.

- *Balanced ER* (D3): two separate backward passes whose gradients are individually unit-normalised before being combined 50/50, removing the magnitude asymmetry between g_{new} and g_{replay} .
- *Full-data replay* (D2, D4): the replay gradient is computed on *every* stored sample exactly once per step (no sub-sampling). Paired with a buffer equal to the full past-task training set (60,000 for rot-MNIST T_0), this yields the deterministic empirical past-task gradient at the current parameters, removing estimator variance.

The λ -curriculum re-weights only the current-task term, minimising $L_\lambda = L_{\text{replay}} + \lambda(t) L_{\text{current}}$; the adaptive schedule reads $\|g_{\text{replay}}\|/\|g_{\text{new}}\|$ through an exponential moving average ($\alpha = 0.05$) and applies the optional floor λ_{min} .

A.4 Asymmetric PER

The filtered push $\delta (F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}}$ of Equation (1.2) is computed each step by warm-started conjugate gradients over Fisher-vector products (25 iterations), so the metric is never formed as a matrix; warm-starting from the previous step’s solution keeps the per-step solve cheap across the run. The solver control of Section 5.2.4 reruns the strongest filter ($\delta = 0.03$, full buffer) at 60 iterations to separate properties of the gate from artefacts of an under-converged solve. On the full-buffer leg the replay gradient is exact over all 60k past samples while the Fisher stays estimated on a sampled batch, since a full-buffer Fisher-vector product per CG iteration would dominate the runtime.

B Metric Definitions

This appendix gives the full definitions of every metric recorded by the evaluation pipeline of Section 5.1.2. The main text explains the measures the argument turns on; the tables below are the look-up reference for the complete set.

Stability-gap metrics. Computed from the dense per-step evaluation. Write a_j^{pre} for the pre-task accuracy on task j and $a_j(t)$ for its accuracy at step t of the new task.

Metric	Definition	Role
<code>gap_depth</code>	$\max_j (a_j^{\text{pre}} - \min_t a_j(t))$ over the full record window	Primary metric, worst single point of instability
<code>gap_area_end</code>	$\sum_j \int_0^T \max(0, b_j - a_j(t)) dt$, trapezoidal over the full new-task record (to task end T), with $b_j = \min(a_j^{\text{pre}}, \bar{a}_j^{\text{end}})$ and $\bar{a}_j^{\text{end}}$ the trailing-window mean (last $\max(5, \lceil 0.1n \rceil)$ samples); accuracy at or above b_j contributes 0	Transient cost reported in the results: dip below the recovered level until it closes, excluding permanent forgetting and continued learning (units: accuracy · steps)
<code>recovery_steps</code>	Given a record has dipped below the threshold, the first step (counted from the transition) at which <i>every</i> past task is simultaneously at $\geq 90\%$ of its pre-task baseline	None if no sub-threshold dip occurs, or if recovery is not reached within the record

Table B.1: Stability-gap metrics, recorded from dense per-step evaluation. `gap_depth` and `gap_area_end` are the two axes used in the main analysis, how deep the gap cuts and how long it stays open, together with `recovery_steps`.

Continual-learning metrics. Following Lange et al. [16], computed at task boundaries and over the full record. Let $A(E_j, f_n)$ be the accuracy on the test set of task j evaluated at the model f_n after training stage n , and let N be the number of tasks. The full task-boundary accuracy matrix $R[i, j] = A(E_j, f_{T_i})$ is logged for every run, so any downstream metric can be reconstructed.

Metric	Definition	Role
ACC	$\frac{1}{N} \sum_j A(E_j, f_{T_{N-1}})$	Average task accuracy at the final model
FORG	$\frac{1}{N-1} \sum_{i < N-1} [A(E_i, f_{T_i}) - A(E_i, f_{T_{N-1}})]$	Average drop from right-after-learning to final
min-ACC	$\frac{1}{N-1} \sum_{i < N-1} \min\{A(E_i, f_n) : n > T_i\}$	Worst-case retention since learning
WF _w	Mean over tasks of the largest accuracy drop in any window of w consecutive evaluations ($w \in \{10, 100\}$)	Worst-case stability over short / medium horizons
WP _w	Symmetric counterpart of WF _w : the largest accuracy gain ($w \in \{10, 100\}$)	Plasticity / learning velocity
WC-ACC	$\frac{1}{N} A(E_{N-1}, f_{T_{N-1}}) + (1 - \frac{1}{N}) \text{min-ACC}$	Worst-case trade-off — a lower bound on ACC

Table B.2: Continual-learning metrics of Lange et al. [16], recorded at task boundaries and over the full record.

Gradient-fidelity diagnostics. Recorded at the dense cadence for the decomposition conditions only. The tracker compares the replay-buffer gradient g_{replay} against the true past-task gradient g_{true} , computed at every step by a separate forward and backward pass over the entire past-task training set (no sub-sampling, so g_{true} is the deterministic empirical past-task gradient at the current parameters).

Diagnostic	Logged name	Role
$\cos(g_{\text{replay}}, g_{\text{true}})$	<code>true_grad_cosine</code>	Buffer-side directional fidelity, the estimator-variance driver of Section 5.2.1
$\ g_{\text{replay}}\ / \ g_{\text{true}}\ $	<code>true_grad_mag_ratio</code>	Buffer-side magnitude fidelity
$\ g_{\text{new}}\ / \ g_{\text{replay}}\ $	<code>grad_ratio</code>	Update-side magnitude asymmetry, large at step 1 when $\ g_{\text{replay}}\ \approx 0$

Table B.3: Gradient-fidelity diagnostics. Each is logged as a per-step series together with its mean (and minimum, for the cosine). The first two carry the `true_grad` prefix because they are measured against g_{true} ; `grad_ratio` compares g_{replay} to the current-task gradient g_{new} and its inverse drives the adaptive curriculum of Section 5.2.2.

C Complete Results

This appendix gives the full metric set for every condition of the main experimental blocks, as recorded by the aggregation pipeline of Section 5.1.2. Each row is the mean over five seeds (three for the CIFAR generalisation block) with the seed standard deviation. Accuracy-type metrics (ACC, FORG, min-ACC, WF₁₀₀, WP₁₀₀, WC-ACC) are fractions; `gap_depth` is in percentage points (pp); `gap_area_end` is the transient cost (accuracy-fraction $\times$ steps, Table B.1), the dip below the level each task recovers to, so permanent forgetting does not contribute; “Recov.” is `recovery_steps`, with “—” marking conditions where every past task stayed above the 90% recovery threshold for the whole window (no recovery event to time).

C.1 rot-MNIST: Decomposition (D-series)

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end	Recov.
<i>Momentum 0.0</i>									
D1 vanilla	0.873 $\pm$ 0.005	0.019 $\pm$ 0.020	0.706 $\pm$ 0.056	0.147 $\pm$ 0.029	0.736 $\pm$ 0.020	0.794 $\pm$ 0.028	19.5 $\pm$ 4.1	6.3 $\pm$ 0.5	3.6 $\pm$ 0.9
D2 fulldata	0.890 $\pm$ 0.009	-0.016 $\pm$ 0.009	0.719 $\pm$ 0.039	0.126 $\pm$ 0.020	0.736 $\pm$ 0.018	0.801 $\pm$ 0.022	18.3 $\pm$ 2.5	8.1 $\pm$ 0.7	3.0 $\pm$ 1.2
D3 balanced	0.849 $\pm$ 0.005	0.031 $\pm$ 0.018	0.824 $\pm$ 0.006	0.045 $\pm$ 0.002	0.721 $\pm$ 0.004	0.836 $\pm$ 0.004	5.7 $\pm$ 1.9	1.4 $\pm$ 0.4	—
D4 fulldata balanced	0.869 $\pm$ 0.002	-0.014 $\pm$ 0.013	0.862 $\pm$ 0.002	0.030 $\pm$ 0.007	0.710 $\pm$ 0.003	0.852 $\pm$ 0.002	1.9 $\pm$ 1.3	3.2 $\pm$ 0.5	—
<i>Momentum 0.9</i>									
D1 vanilla	0.928 $\pm$ 0.003	0.058 $\pm$ 0.007	0.797 $\pm$ 0.028	0.092 $\pm$ 0.016	0.808 $\pm$ 0.012	0.878 $\pm$ 0.014	15.9 $\pm$ 2.6	1.0 $\pm$ 0.2	12.6 $\pm$ 1.7
D2 fulldata	0.964 $\pm$ 0.002	-0.010 $\pm$ 0.003	0.807 $\pm$ 0.009	0.082 $\pm$ 0.004	0.808 $\pm$ 0.012	0.885 $\pm$ 0.004	14.9 $\pm$ 0.9	3.1 $\pm$ 0.2	10.6 $\pm$ 1.1
D3 balanced	0.932 $\pm$ 0.005	0.038 $\pm$ 0.003	0.897 $\pm$ 0.006	0.040 $\pm$ 0.003	0.822 $\pm$ 0.011	0.922 $\pm$ 0.003	5.9 $\pm$ 0.6	0.4 $\pm$ 0.2	—
D4 fulldata balanced	0.967 $\pm$ 0.001	-0.022 $\pm$ 0.003	0.942 $\pm$ 0.005	0.016 $\pm$ 0.004	0.834 $\pm$ 0.010	0.950 $\pm$ 0.002	1.3 $\pm$ 0.4	0.2 $\pm$ 0.1	—

Table C.1: rot-MNIST decomposition block, full metrics, at both momentum settings. Buffer-fidelity diagnostics for these runs are in Table C.4.

C.2 rot-MNIST: λ -Curriculum (C-series)

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end	Recov.
<i>Momentum 0.0</i>									
C1 linear N50	0.851 $\pm$ 0.023	0.042 $\pm$ 0.021	0.707 $\pm$ 0.019	0.148 $\pm$ 0.011	0.748 $\pm$ 0.006	0.784 $\pm$ 0.021	17.5 $\pm$ 1.7	4.9 $\pm$ 0.5	35.8 $\pm$ 6.3
C2 linear N100	0.844 $\pm$ 0.025	0.046 $\pm$ 0.023	0.719 $\pm$ 0.033	0.140 $\pm$ 0.025	0.730 $\pm$ 0.006	0.786 $\pm$ 0.019	16.2 $\pm$ 4.3	3.6 $\pm$ 0.3	64.0 $\pm$ 8.6
C3 linear N200	0.828 $\pm$ 0.025	0.054 $\pm$ 0.024	0.739 $\pm$ 0.048	0.128 $\pm$ 0.043	0.688 $\pm$ 0.004	0.784 $\pm$ 0.034	14.3 $\pm$ 4.6	1.8 $\pm$ 0.6	132.8 $\pm$ 43.4
C4 adaptive	0.833 $\pm$ 0.025	0.050 $\pm$ 0.019	0.754 $\pm$ 0.025	0.111 $\pm$ 0.016	0.703 $\pm$ 0.007	0.794 $\pm$ 0.020	12.8 $\pm$ 2.9	2.2 $\pm$ 0.5	88.8 $\pm$ 6.1
C5 adaptive lmin0.05	0.835 $\pm$ 0.025	0.048 $\pm$ 0.019	0.749 $\pm$ 0.027	0.113 $\pm$ 0.016	0.702 $\pm$ 0.006	0.793 $\pm$ 0.020	13.2 $\pm$ 3.2	2.2 $\pm$ 0.5	88.8 $\pm$ 6.1
C6 adaptive lmin0.10	0.836 $\pm$ 0.026	0.048 $\pm$ 0.021	0.743 $\pm$ 0.026	0.120 $\pm$ 0.019	0.700 $\pm$ 0.007	0.791 $\pm$ 0.019	13.9 $\pm$ 3.3	2.4 $\pm$ 0.5	88.4 $\pm$ 11.0
C7 adaptive lmin0.20	0.837 $\pm$ 0.031	0.049 $\pm$ 0.024	0.743 $\pm$ 0.030	0.119 $\pm$ 0.023	0.710 $\pm$ 0.010	0.792 $\pm$ 0.024	13.9 $\pm$ 3.4	2.7 $\pm$ 0.6	64.8 $\pm$ 8.2
<i>Momentum 0.9</i>									
C1 linear N50	0.932 $\pm$ 0.003	0.053 $\pm$ 0.005	0.892 $\pm$ 0.004	0.041 $\pm$ 0.004	0.830 $\pm$ 0.012	0.927 $\pm$ 0.003	6.4 $\pm$ 0.5	0.2 $\pm$ 0.1	—
C2 linear N100	0.932 $\pm$ 0.003	0.051 $\pm$ 0.008	0.892 $\pm$ 0.005	0.036 $\pm$ 0.003	0.826 $\pm$ 0.012	0.926 $\pm$ 0.002	6.4 $\pm$ 0.6	0.2 $\pm$ 0.1	—
C3 linear N200	0.925 $\pm$ 0.007	0.051 $\pm$ 0.006	0.893 $\pm$ 0.004	0.031 $\pm$ 0.004	0.819 $\pm$ 0.010	0.920 $\pm$ 0.006	6.2 $\pm$ 0.6	0.1 $\pm$ 0.0	—
C4 adaptive	0.917 $\pm$ 0.001	0.019 $\pm$ 0.004	0.931 $\pm$ 0.005	0.016 $\pm$ 0.003	0.798 $\pm$ 0.010	0.915 $\pm$ 0.001	2.5 $\pm$ 0.5	0.3 $\pm$ 0.1	—
C5 adaptive lmin0.05	0.922 $\pm$ 0.001	0.023 $\pm$ 0.004	0.930 $\pm$ 0.005	0.016 $\pm$ 0.002	0.802 $\pm$ 0.009	0.921 $\pm$ 0.001	2.6 $\pm$ 0.5	0.1 $\pm$ 0.0	—
C6 adaptive lmin0.10	0.927 $\pm$ 0.002	0.028 $\pm$ 0.005	0.926 $\pm$ 0.004	0.017 $\pm$ 0.003	0.806 $\pm$ 0.009	0.927 $\pm$ 0.002	3.0 $\pm$ 0.4	0.0 $\pm$ 0.0	—
C7 adaptive lmin0.20	0.931 $\pm$ 0.002	0.035 $\pm$ 0.005	0.919 $\pm$ 0.003	0.021 $\pm$ 0.004	0.812 $\pm$ 0.009	0.930 $\pm$ 0.001	3.7 $\pm$ 0.4	0.0 $\pm$ 0.0	—

Table C.2: rot-MNIST λ -curriculum sweep, full metrics, at both momentum settings. All conditions applied on top of standard ER (1k reservoir).

Full-buffer momentum diagnostic (C8). Table C.3 crosses the practical curriculum with the full 60k buffer at both momentum settings, isolating momentum’s two roles (Section 6.2). Source variance reduction alone (C8, full buffer, $\mu=0$) already closes the gap to 0.2 pp, so the residual the curriculum leaves at $\mu=0$ is estimator variance, not structure; momentum’s gap-closing role is interchangeable with the full buffer. Acceleration alone (the $\mu=0.9$ column) restores accuracy: C8_M reaches 0.955, approaching the full-data ceiling (D2_M: 0.964), while re-introducing only a 2.0 pp transient. C8_M’s tail (area 0.38) exceeds C7_M’s (0.03) because C7_M’s residual is a permanent step to a lower plateau that the end-referenced area excludes (Section 5.1.2), whereas C8_M recovers to a higher level, so its dip is a genuine transient the area counts in full. C8/C8_M are diagnostic probes: the full buffer defeats the limited-memory premise, and the practical scalar-gate configuration remains C7_M (3.7 pp / 0.03 / 0.931) at 1/60th the memory.

	C7 (1k, $\mu=0$)	C7 _M (1k, $\mu=0.9$)	C8 (full, $\mu=0$)	C8 _M (full, $\mu=0.9$)
gap_depth (pp)	13.9	3.7	0.2	2.0
gap_area_end	2.7	0.03	0.05	0.38
ACC	0.837	0.931	0.818	0.955

Table C.3: Curriculum $\times$ full buffer $\times$ momentum on rot-MNIST, isolating momentum’s two roles. Source variance reduction alone (C8) closes the gap; acceleration alone (the μ column) restores accuracy. The residual at C7 ($\mu=0$) is variance, not structure.

C.3 rot-MNIST: Buffer-Fidelity Diagnostics

The gradient diagnostics of Appendix B were logged for the decomposition block only. The full-data conditions (D2, D4) reach exact fidelity by construction, the replay gradient equals the deterministic empirical past-task gradient, so cosine and magnitude ratio are 1.000. The finite-buffer conditions (D1, D3) show the directional and magnitude error a 1,000-sample reservoir incurs against g_{true} .

Condition	cōs	cos _{min}	$\bar{\rho}$
D1 vanilla ($\mu=0$)	0.711 $\pm$ 0.035	0.086 $\pm$ 0.103	1.040 $\pm$ 0.039
D3 balanced ($\mu=0$)	0.665 $\pm$ 0.043	0.025 $\pm$ 0.140	1.137 $\pm$ 0.060
D1 vanilla ($\mu=0.9$)	0.551 $\pm$ 0.033	0.135 $\pm$ 0.069	0.418 $\pm$ 0.014
D3 balanced ($\mu=0.9$)	0.441 $\pm$ 0.039	-0.052 $\pm$ 0.117	0.265 $\pm$ 0.024
D2, D4 (full data)	1.000	1.000	1.000

Table C.4: Buffer-fidelity diagnostics for the decomposition block: cōs and cos_{min} are the mean and minimum of $\cos(g_{\text{replay}}, g_{\text{true}})$ over the window; $\bar{\rho}$ is the mean of $\|g_{\text{replay}}\|/\|g_{\text{true}}\|$.

C.4 rot-MNIST: Asymmetric PER (P-series)

Table C.5 gives the full metric set for the asymmetric-PER δ sweep on both buffer legs and at both momentum settings; the momentum-0 depth/area/ACC columns match Table 6.2 in the main text, and the momentum-0.9 rows are the source of the re-inflation numbers of Section 6.3. Momentum-0.9 runs cover the three strongest filters on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg.

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
<i>Standard (1k) buffer, momentum 0.0</i>								
$\delta=1.0$	0.869 ± 0.005	0.027 ± 0.018	0.792 ± 0.012	0.081 ± 0.009	0.733 ± 0.009	0.838 ± 0.006	9.0 ± 0.6	3.29 ± 0.65
$\delta=0.3$	0.873 ± 0.004	0.023 ± 0.017	0.807 ± 0.012	0.068 ± 0.009	0.736 ± 0.008	0.847 ± 0.005	7.4 ± 1.1	2.32 ± 0.59
$\delta=0.1$	0.876 ± 0.004	0.018 ± 0.016	0.822 ± 0.012	0.056 ± 0.009	0.739 ± 0.008	0.856 ± 0.005	5.9 ± 1.5	1.56 ± 0.65
$\delta=0.03$	0.880 ± 0.005	0.012 ± 0.015	0.834 ± 0.012	0.048 ± 0.005	0.741 ± 0.007	0.862 ± 0.005	4.8 ± 1.9	1.04 ± 0.58
<i>Standard (1k) buffer, momentum 0.9</i>								
$\delta=0.3$	0.929 ± 0.002	0.061 ± 0.004	0.851 ± 0.008	0.065 ± 0.004	0.813 ± 0.009	0.907 ± 0.005	10.4 ± 0.5	0.50 ± 0.17
$\delta=0.1$	0.928 ± 0.002	0.061 ± 0.006	0.861 ± 0.008	0.060 ± 0.005	0.815 ± 0.009	0.911 ± 0.005	9.4 ± 0.5	0.38 ± 0.19
$\delta=0.03$	0.929 ± 0.003	0.060 ± 0.009	0.871 ± 0.008	0.053 ± 0.007	0.816 ± 0.010	0.917 ± 0.005	8.5 ± 0.6	0.23 ± 0.12
<i>Full (60k) buffer, momentum 0.0</i>								
$\delta=1.0$	0.898 ± 0.002	-0.025 ± 0.012	0.841 ± 0.011	0.037 ± 0.011	0.734 ± 0.006	0.865 ± 0.006	4.0 ± 1.0	2.32 ± 1.38
$\delta=0.3$	0.900 ± 0.002	-0.029 ± 0.012	0.867 ± 0.002	0.026 ± 0.007	0.739 ± 0.005	0.878 ± 0.002	1.5 ± 1.1	1.01 ± 0.91
$\delta=0.1$	0.901 ± 0.002	-0.032 ± 0.013	0.860 ± 0.026	0.040 ± 0.028	0.741 ± 0.004	0.875 ± 0.013	2.2 ± 1.6	0.44 ± 0.38
$\delta=0.03$	0.903 ± 0.002	-0.035 ± 0.012	0.787 ± 0.094	0.118 ± 0.102	0.742 ± 0.004	0.838 ± 0.048	9.5 ± 8.2	0.43 ± 0.25
$\delta=0.03$ (CG 60)	0.903 ± 0.002	-0.035 ± 0.012	0.766 ± 0.113	0.126 ± 0.109	0.742 ± 0.004	0.828 ± 0.057	11.5 ± 10.1	0.47 ± 0.28
<i>Full (60k) buffer, momentum 0.9</i>								
$\delta=0.3$	0.964 ± 0.001	-0.012 ± 0.002	0.864 ± 0.009	0.052 ± 0.005	0.813 ± 0.008	0.913 ± 0.005	9.1 ± 0.8	1.91 ± 0.26
$\delta=0.1$	0.965 ± 0.001	-0.012 ± 0.003	0.880 ± 0.008	0.044 ± 0.004	0.814 ± 0.008	0.921 ± 0.005	7.6 ± 0.7	1.57 ± 0.25

Table C.5: rot-MNIST asymmetric-PER δ sweep, full metrics: both buffer legs at both momentum settings, plus the solver control (CG 60). The momentum-0.9 rows show the re-inflation of Section 6.3: depth climbs on every leg relative to its momentum-0 counterpart, while the full-buffer accuracies (≈ 0.965) are diagnostic values only, the full buffer defeating the limited-memory premise.

C.5 rot-MNIST: Symmetric vs. Asymmetric Preconditioning

Leaving the restoring gradient g_{rep} at full strength is the directional gate’s defining choice (Section 6.3). The alternative is to precondition the summed update, $d = \delta(F_{\text{rep}} + \delta I)^{-1}(g_{\text{cur}} + g_{\text{rep}})$, the symmetric variant that passes both gradients through the same metric. Because g_{rep} is the gradient of the replay loss, it lies in the top eigenspace of that loss’s own Fisher F_{rep} , so the symmetric filter shrinks the restoring gradient *harder* than the harmful push it is meant to brake and cannot preferentially protect the past task. Table C.6 sweeps the same δ on both variants, and they are mirror images. As the filter strengthens ($\delta \rightarrow 0$) the asymmetric gate’s forgetting falls ($2.7 \rightarrow 1.2$ pp) and its accuracy rises, while the symmetric gate’s forgetting *climbs* ($3.0 \rightarrow 6.6$ pp) and its accuracy falls; every one of these four trends holds across all five seeds. Symmetric depth barely moves across the sweep ($5.7 \rightarrow 6.7$ pp) and at $\delta=1.0$ even sits below the asymmetric gate’s (5.7 vs. 9.0 pp), because filtering the whole update also damps the per-step *expression* of the dip. This is the area caveat of Section 5.1.2 made concrete: the symmetric gate’s end-referenced area collapses toward zero ($2.7 \rightarrow 0.01$) not because the dip recovers but because it stops recovering, settling into a permanent accommodation that the area excludes and the forgetting metric records. Read on depth or transient area the symmetric gate looks competitive; read on forgetting and final accuracy it degrades monotonically as its filter strengthens. Asymmetry is what turns the filter from a reshuffling of the damage into a net removal of it.

δ	Asymmetric (g_{rep} free)			Symmetric (joint)		
	depth (pp)	FORG (pp)	ACC	depth (pp)	FORG (pp)	ACC
1.0	9.0	2.7	0.869	5.7	3.0	0.872
0.3	7.4	2.3	0.873	5.9	4.2	0.867
0.1	5.9	1.8	0.876	6.2	5.7	0.861
0.03	4.8	1.2	0.880	6.7	6.6	0.857

Table C.6: Asymmetric versus symmetric preconditioning on the standard 1k buffer at momentum 0, five seeds, sweeping the shared damping δ . Both variants filter through the replay Fisher F_{rep} ; they differ only in whether the metric is applied to the new-task push alone (asymmetric, g_{rep} left free) or to the summed update (symmetric). As $\delta \rightarrow 0$ the asymmetric gate improves on every axis while the symmetric gate degrades on every axis; symmetric depth stays flat because the filter suppresses the per-step expression of the dip even as the underlying forgetting grows (Section 5.1.2).

C.6 CIFAR-10: Headline Block

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
G1 vanilla BN	0.723 ± 0.008	0.010 ± 0.019	0.644 ± 0.023	0.127 ± 0.015	0.484 ± 0.020	0.667 ± 0.016	11.8 ± 2.5	17.5 ± 4.6
G2 adaptive BN	0.722 ± 0.011	0.006 ± 0.016	0.713 ± 0.017	0.070 ± 0.010	0.469 ± 0.016	0.698 ± 0.013	4.8 ± 0.7	3.6 ± 2.2
G3 asym. PER BN	0.715 ± 0.011	-0.008 ± 0.018	0.698 ± 0.023	0.124 ± 0.022	0.460 ± 0.025	0.682 ± 0.023	5.9 ± 4.3	3.3 ± 3.0
G4 vanilla GN	0.730 ± 0.011	-0.023 ± 0.030	0.309 ± 0.111	0.294 ± 0.097	0.415 ± 0.058	0.507 ± 0.054	41.2 ± 13.5	34.9 ± 23.7
G5 adaptive GN	0.731 ± 0.020	-0.026 ± 0.026	0.670 ± 0.055	0.095 ± 0.033	0.340 ± 0.052	0.687 ± 0.037	5.2 ± 2.1	1.7 ± 0.9
G6 asym. PER GN	0.741 ± 0.009	-0.035 ± 0.018	0.673 ± 0.023	0.080 ± 0.009	0.330 ± 0.024	0.696 ± 0.015	5.6 ± 1.1	4.8 ± 3.8

Table C.7: CIFAR-10 headline block, full metrics (ResNet-18, momentum 0.9, buffer 5,000, five seeds). “adaptive” is the practical curriculum with $\lambda_{\min} = 0.20$; “asym. PER” runs at $\delta = 0.1$.

C.7 CIFAR-10: Generalisation Block

Each shift pairs the practical curriculum (“curric.”: adaptive $\lambda_{\min} = 0.20$, buffer 5,000) against a vanilla-ER control on the same shift, at the indicated normalisation, over three seeds.

Condition	ACC	FORG	min-ACC	WC-ACC	Depth	Area end
<i>BatchNorm</i>						
shot noise — vanilla	0.724 ± 0.011	0.017 ± 0.010	0.633 ± 0.017	0.663 ± 0.014	13.1 ± 3.0	21.2 ± 3.9
shot noise — curric.	0.723 ± 0.014	0.010 ± 0.010	0.708 ± 0.032	0.697 ± 0.016	5.6 ± 1.0	5.8 ± 1.5
contrast — vanilla	0.671 ± 0.026	-0.008 ± 0.006	0.610 ± 0.063	0.586 ± 0.048	15.4 ± 3.9	33.4 ± 4.5
contrast — curric.	0.671 ± 0.044	-0.012 ± 0.007	0.727 ± 0.034	0.643 ± 0.048	3.7 ± 1.1	2.2 ± 0.9
3-task — vanilla	0.740 ± 0.006	-0.011 ± 0.013	0.670 ± 0.015	0.691 ± 0.010	3.3 ± 0.4	6.0 ± 0.7
3-task — curric.	0.745 ± 0.010	-0.021 ± 0.008	0.677 ± 0.025	0.696 ± 0.017	6.9 ± 2.7	3.9 ± 0.1
<i>GroupNorm</i>						
shot noise — vanilla	0.718 ± 0.021	-0.036 ± 0.036	0.244 ± 0.075	0.469 ± 0.032	46.3 ± 12.0	54.4 ± 40.8
shot noise — curric.	0.705 ± 0.061	-0.028 ± 0.012	0.638 ± 0.108	0.656 ± 0.087	7.3 ± 5.9	2.4 ± 1.1
contrast — vanilla	0.777 ± 0.022	-0.067 ± 0.021	0.641 ± 0.061	0.710 ± 0.040	8.4 ± 4.4	3.3 ± 0.6
contrast — curric.	0.778 ± 0.032	-0.070 ± 0.013	0.668 ± 0.077	0.724 ± 0.054	3.9 ± 3.1	1.2 ± 0.8
3-task — vanilla	0.722 ± 0.020	-0.023 ± 0.015	0.468 ± 0.043	0.550 ± 0.023	4.4 ± 0.4	10.3 ± 3.6
3-task — curric.	0.721 ± 0.019	-0.025 ± 0.020	0.421 ± 0.114	0.517 ± 0.070	32.8 ± 14.5	32.0 ± 19.5

Table C.8: CIFAR-10 generalisation block, full metrics. The two-task novel corruptions (shot noise, contrast) favour the curriculum under both normalisations; the three-task group-norm sequence is the outlier analysed below.

The three-task group-norm failure. The three-task sequence is the one place the curriculum underperforms its control (depth 32.8 ± 14.5 pp against vanilla’s 4.4 ± 0.4 , Table C.8; Figure C.3), and the logged gradient-norm traces locate the failure in the conditioning of the feedback signal. The adaptive ratio $r = \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ carries information through the boundary ordering of Section 5.2.2: at a well-separated switch the new-task gradient dwarfs the replay gradient, so r starts near zero and its rise tracks the replay gradient reasserting itself. The first boundary of this sequence behaves that way (clean $\rightarrow$ Gaussian: first-step new-task loss 1.9–2.1, $r \approx 0.2$), and the schedule holds every past task flat through it. The second boundary never establishes the ordering: Gaussian and shot noise are near-identical, so the incoming model already fits the new task (T_2 accuracy 0.67–0.71 before its first step, new-task loss 0.5–0.7), leaving $\|g_{\text{new}}\|$ small, while $\|g_{\text{replay}}\|$ is small because the model is converged on the replayed tasks. The ratio becomes a quotient of two near-zero norms: under group norm it reads 0.5–1.3 from the first step and fluctuates step to step, so the EMA drives λ from the floor to ≈ 0.7 –0.8 within fifty steps and back to $\lambda_{\min}$ within a few hundred. The gate is moved by noise, not by damage.

What follows, in all three seeds, is not a stability gap but an optimisation instability. Within the first forty steps the replay loss itself diverges (0.2 – $0.4 \rightarrow 0.8$ – 2.1) and every task falls together, *including the one being trained* (T_2 : $0.71 \rightarrow 0.29$, $0.69 \rightarrow 0.30$, $0.67 \rightarrow 0.51$ across seeds), before the run recovers; the per-seed depth scales with the divergence, which is where the ± 14.5 pp spread comes from. Because all tasks crash together, the depth and area of the three-task group-norm rows record this instability, not the past-task transient the rest of the study measures. Two controls isolate the trigger. The vanilla run crosses the same boundary with a flat replay loss and 4.4 pp of depth, so the near-identical boundary is harmless on its own; the batch-norm run executes the identical schedule through the same boundary, with the same ill-conditioned early ratio, without instability (Figure C.3), so the noise-driven λ is not sufficient either. The failure needs both the unconditioned signal and the group-norm backbone’s boundary sensitivity, consistent with the far deeper group-norm transients vanilla ER shows on the same corruption family (shot noise: 46.3 vs 13.1 pp, Table C.8). We therefore read the episode as a reliability boundary of the gradient-norm-ratio signal rather than of the gate principle: the ratio loses its meaning when both norms become small, and the design admits direct guards — freezing λ when both norms fall below a reference scale, or reading a damage signal that does not vanish at a quiet boundary, such as the replay-loss excess over its pre-switch level.

C.8 CIFAR-10: Momentum Cross and Compute Cost

This subsection collects the three deep-backbone checks summarised in Section 6.5: the momentum cross (Table C.9), the generalisation block (Table C.8 above), and the compute cost (Table C.10).

Momentum cross. The cross repeats the composability test of Section 5.2.3 on the ResNet, pairing momentum-0 runs at group norm against the momentum-0.9 rows. Two of the three predicted signatures reproduce: momentum leaves vanilla ER’s depth essentially unchanged ($40.5 \rightarrow 41.2$ pp, the deterministic-first-step null), and it composes with the curriculum, taking depth from 28.6 to 5.2 pp (source attenuation). The third, momentum re-inflating the directional gate, does not appear: momentum takes asymmetric PER from 22.6 to 5.6 pp instead. We read this as a convergence confound rather than a failure of the account: at momentum 0 the ResNet under-converges the past task in ten epochs (vanilla min-ACC 0.11, ACC 0.65), so the momentum-0 depths sit on a degraded baseline and the directional-gate contrast is dominated by the general convergence gain rather than the per-step-attenuation effect the full-buffer MLP isolates cleanly (Section 6.3). On this backbone we confirm the mechanism only for the two gates whose momentum contrast is not swamped by convergence.

Method (GroupNorm)	depth, $\mu=0$	depth, $\mu=0.9$
Vanilla ER	40.5 ± 4.0	41.2 ± 13.5
Curriculum ($\lambda_{\min}=0.20$)	28.6 ± 8.8	5.2 ± 2.1
Asym. PER ($\delta=0.1$)	22.6 ± 9.7	5.6 ± 1.1

Table C.9: CIFAR-10 momentum cross (group norm, buffer 5,000, five seeds), gap depth in pp. The vanilla null and the curriculum composition reproduce; the directional-gate re-inflation does not, confounded by momentum-0 under-convergence of the ResNet.

Compute cost. Cost is measured with a dedicated benchmark: one full two-task run per gate in its headline configuration (curriculum: adaptive schedule, $\lambda_{\min}=0.20$; asymmetric PER: $\delta=0.1$, 25 warm-started CG iterations; momentum 0.9, group norm and buffer 5,000 on CIFAR), five seeds each on a single NVIDIA RTX PRO 6000, with per-step stability evaluation disabled so the numbers reflect method compute alone. Wall-clock and CPU time come from `getrusage`; GPU utilisation, power, and memory are sampled at 1 Hz with `nvidia-smi`, and energy is the integrated power. On the ResNet-18 the split is stark (Table C.10). The scalar gate adds one gradient-norm ratio and an EMA update to a standard ER step, and its run is not even compute-bound: the GPU sits half-idle at 49% utilisation and 259 W. The directional gate’s 25 conjugate-gradient Fisher-vector products per step, each on the order of a backward pass, saturate the same card (96%, 367 W) for $16\times$ as long, and since energy is power times time the two factors compound to a $24\times$ energy gap (13 vs. 320 Wh per run). Memory is not the price: peak GPU memory differs by 11%. On the rot-MNIST MLP both gates finish in under half a minute at near-idle utilisation — the wall-clock is fixed startup, not method compute — so no meaningful ratio exists there and the cost question is decided entirely on the deep backbone. The ratios are the transferable claim; the absolute numbers are specific to this card, and because $\sim 15\text{--}30$ s of each run is startup, the training-only ratio is if anything slightly above $16\times$.

Dataset	Gate	wall [s]	util [%]	power [W]	energy [Wh]	mem [MB]
CIFAR-10 (ResNet-18)	Curriculum (scalar)	198 ± 7	49	259	13	4,203
	Asym. PER ($\delta=0.1$)	$3,157 \pm 11$	96	367	320	4,679
rot-MNIST (MLP)	Curriculum (scalar)	25 ± 1	<1	—	0.4	749
	Asym. PER ($\delta=0.1$)	29 ± 1	8	—	0.5	769

Table C.10: Compute cost of one full two-task run (mean over five seeds, $\pm$ sd on wall-clock; single RTX PRO 6000; per-step evaluation disabled): wall-clock, mean GPU utilisation and power, integrated GPU energy, and peak GPU memory. On the ResNet-18 the directional gate costs $\sim 16\times$ the wall-clock and $\sim 24\times$ the energy of the scalar gate at similar peak memory. The MLP rows are dominated by fixed startup on a near-idle GPU, so their power draw is idle draw (omitted) and no ratio is meaningful.

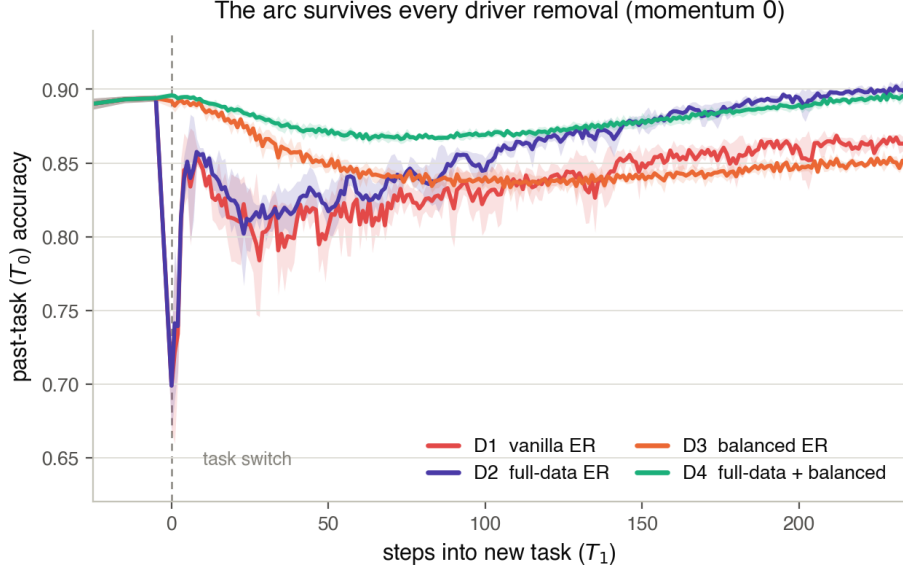


Figure C.1: The full driver ladder: per-step past-task (T_0) accuracy across the $T_0 \rightarrow T_1$ transition for the four driver-removal conditions at momentum 0; solid lines are the seed mean, bands the standard deviation, $x=0$ the task switch. *Depth* of the boundary spike falls when balancing switches off the imbalance driver (deep for the unbalanced D1, D2; shallow for the balanced D3, D4), while *jitter* vanishes when the exact replay gradient switches off the variance driver (the full-data D2, D4 are smooth, the finite-buffer D1, D3 noisy). What no removal touches is the smooth dip-and-recover *bend* beneath the spike, cleanest in the exact-gradient D4 where no jitter can hide it: that bend is the deterministic arc, and it responds to no optimiser hygiene, only to a change of path.

C.9 Long-Sequence rot-MNIST (L-series)

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
<i>Momentum 0.0</i>								
L1 vanilla ER	0.861 ± 0.004	0.060 ± 0.004	0.670 ± 0.055	0.207 ± 0.053	0.443 ± 0.011	0.722 ± 0.043	21.0 ± 5.8	3.14 ± 0.77
L2 curriculum $\lambda_{\min}=0.20$	0.860 ± 0.002	0.031 ± 0.005	0.828 ± 0.007	0.073 ± 0.005	0.400 ± 0.003	0.840 ± 0.005	4.1 ± 0.8	1.62 ± 0.42
L3 asym. PER $\delta=0.1$	0.870 ± 0.003	0.053 ± 0.006	0.843 ± 0.008	0.038 ± 0.007	0.381 ± 0.003	0.862 ± 0.007	4.7 ± 1.2	0.38 ± 0.24
<i>Momentum 0.9</i>								
L1 vanilla ER	0.889 ± 0.007	0.096 ± 0.008	0.845 ± 0.007	0.065 ± 0.008	0.317 ± 0.006	0.870 ± 0.006	8.9 ± 1.7	1.68 ± 0.69
L2 curriculum $\lambda_{\min}=0.20$	0.897 ± 0.002	0.071 ± 0.003	0.871 ± 0.005	0.041 ± 0.004	0.364 ± 0.005	0.888 ± 0.004	5.4 ± 0.6	0.60 ± 0.14
L3 asym. PER $\delta=0.1$	0.884 ± 0.004	0.104 ± 0.004	0.849 ± 0.006	0.054 ± 0.004	0.325 ± 0.005	0.873 ± 0.005	7.5 ± 1.5	0.80 ± 0.63

Table C.11: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive transitions, five seeds), the L-series: both gates in their practical configurations against vanilla ER, at both momentum settings. Depth is the worst single-boundary drop (gap_depth) and area the end-referenced transient (gap_area_end), aggregated as in the two-task blocks so the numbers are directly comparable. At momentum 0 both gates cut the depth from 21 pp to ≈ 4 pp; under momentum, where the per-boundary drops are already smaller, the curriculum stays ahead on cumulative retention (lowest FORG, highest ACC and WC-ACC) while asymmetric PER holds the smallest per-boundary area but, its per-step filter re-inflated by momentum (Section 6.3), gives up its retention edge, falling slightly behind vanilla on FORG (0.104 vs. 0.096, all five seeds).

C.10 Supplementary Figures

This subsection collects supplementary figures: the full driver ladder (Figure C.1), the five-task rot-MNIST sequence (Figure C.2, reported in Table C.11), and the three-task group-norm generalisation outlier (Figure C.3) analysed in Appendix C.7.

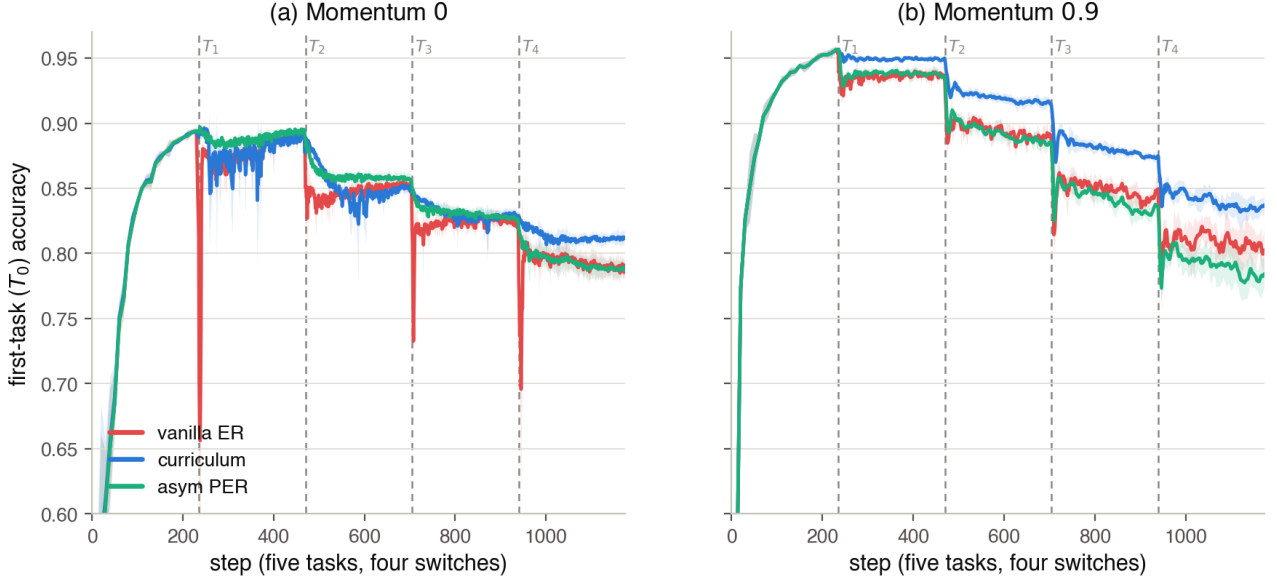


Figure C.2: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive switches, five seeds with $\pm$ std bands), the L-series: first-task T_0 accuracy across the whole sequence for vanilla ER (red), the practical curriculum $\lambda_{\min}=0.20$ (blue) and asymmetric PER $\delta=0.1$ (aqua); dashed lines mark the four switches, and the two panels are the two momentum settings. (a) Momentum 0: both gates hold T_0 almost flat through every boundary — asymmetric PER the smoothest of all, with only shallow dips (smallest per-boundary area, Table C.11) — while vanilla ER takes a sharp instantaneous drop at each switch. This is the near-gap-free directional-gate regime the two-task sweep also finds at momentum 0. (b) Momentum 0.9: momentum re-inflates PER’s per-step filter, exactly as on the two-task benchmark (Section 6.3), so its boundary transients return and its cumulative forgetting falls slightly behind vanilla’s; the curriculum keeps the boundaries smoothest and holds the highest plateau, the least cumulative forgetting on the long horizon. The single-transition reading thus carries past one boundary, and the momentum roles carry with it.

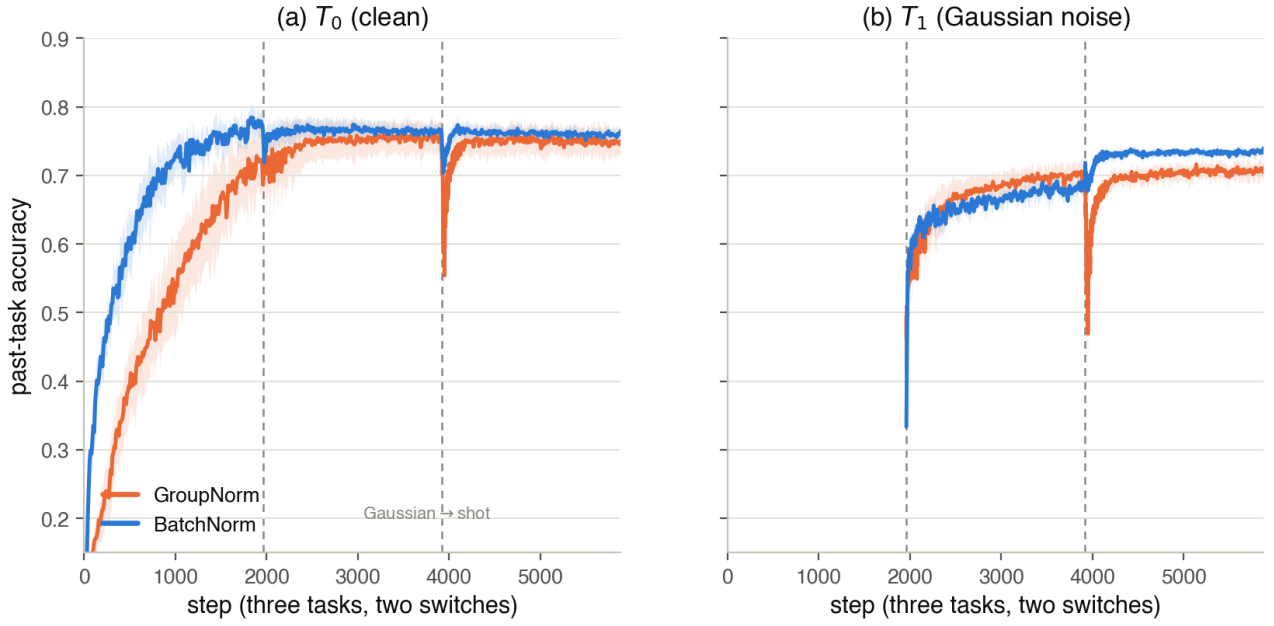


Figure C.3: Per-step past-task accuracy across the three-task CIFAR-10 generalisation sequence (none $\rightarrow$ Gaussian $\rightarrow$ shot, ResNet-18, practical curriculum, three seeds with $\pm$ std bands), comparing the group-norm run (orange) against its batch-norm counterpart (blue). The two switches (dashed) fall at $\sim 2k$ and $\sim 4k$ steps. The first boundary (none $\rightarrow$ Gaussian) barely perturbs the past tasks in either norm. The second (Gaussian $\rightarrow$ shot, two near-identical corruptions) drives a deep transient in T_0 (a) and T_1 (b) under group norm only, while the batch-norm run holds flat through both switches. The dip is thus localised entirely to the second transition and to group norm, the adaptive-ratio reliability boundary analysed in Appendix C.7.

D Detailed Step-by-Step Derivation of the Trajectory Bound

Setup and Definitions

1. The Combined Loss. The total loss function at any point $\lambda \in [0, 1]$ in the curriculum is defined as:

$$L_\lambda(\theta) = L_{\text{replay}}(\theta) + \lambda \cdot L_{\text{current}}(\theta) \quad (\text{D.1})$$

2. Defining the Hessians (∇^2). Differentiating the gradient once more gives the second-derivative matrix, the Hessian (H).

- $H_{\text{replay}} = \nabla^2 L_{\text{replay}}(\theta)$ (The curvature of the old task)
- $H_{\text{current}} = \nabla^2 L_{\text{current}}(\theta)$ (The curvature of the new task)

Therefore, the Hessian of our combined loss, H_λ , is defined by applying the second derivative operator to the total loss function:

$$\begin{aligned} H_\lambda &= \nabla^2 [L_{\text{replay}}(\theta) + \lambda \cdot L_{\text{current}}(\theta)] \\ H_\lambda &= \nabla^2 L_{\text{replay}}(\theta) + \lambda \cdot \nabla^2 L_{\text{current}}(\theta) \\ H_\lambda &= H_{\text{replay}} + \lambda \cdot H_{\text{current}} \end{aligned}$$

3. Assumption: Local-Minimum along the Interpolation Branch. We assume that the loss functions L_{replay} and L_{current} are twice continuously differentiable (C^2) in a neighbourhood of the interpolation path. Furthermore, we assume there exists a continuously differentiable (C^1) branch of critical points $\lambda \mapsto \Theta^*(\lambda)$ for the interpolated loss L_λ , initialized at $\Theta^*(0) = \theta_0^*$, such that the Hessian remains strictly positive-definite: $H_\lambda(\Theta^*(\lambda)) \succ 0$ for every $\lambda \in [0, 1]$.

This assumption ensures that the iterate tracks a continuous family of strict local minima rather than relying on global convexity. Near $\lambda = 0$, this condition is naturally satisfied. Because the initial state θ_0^* is a non-degenerate local minimum of L_{replay} , it follows that $H_0 = H_{\text{replay}}(\theta_0^*) \succ 0$. Consequently, the implicit function theorem guarantees a unique smooth branch in the immediate neighbourhood of $\lambda = 0$.

The persistence of this positive-definite branch as λ increases is bounded by Weyl's inequality applied to $H_\lambda = H_{\text{replay}} + \lambda H_{\text{current}}$; writing $\mu_{\min}(\cdot)$ for the smallest eigenvalue,

$$\mu_{\min}(H_\lambda) \geq \mu_{\min}(H_{\text{replay}}) + \lambda \mu_{\min}(H_{\text{current}}).$$

If the new task is locally convex at these points ($H_{\text{current}} \succeq 0$), positive-definiteness holds for all $\lambda \geq 0$ without requiring a global convexity assumption.

Part A: The Replay Loss Is Monotone Along the Reference Path

Let $\Theta^*(\lambda)$ be the reference path, the exact parameter state that minimises $L_\lambda(\theta)$ for any given λ .

Step 1: The First-Order Condition. Because $\Theta^*(\lambda)$ is a minimum, the gradient of the total loss evaluated at that exact point must be zero:

$$\begin{aligned} \nabla [L_\lambda(\Theta^*(\lambda))] &= 0 \\ \nabla [L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot L_{\text{current}}(\Theta^*(\lambda))] &= 0 \end{aligned}$$

Distributing the gradient operator to both terms gives us the fundamental First-Order Condition:

$$\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0 \quad (\text{D.2})$$

Step 2: Implicit Differentiation to Find Path Movement. We want to know how the reference path Θ^* moves as λ changes. We apply the derivative operator $\frac{d}{d\lambda}$ to the entire First-Order Condition:

$$\frac{d}{d\lambda} [\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))] = \frac{d}{d\lambda} [0]$$

Distributing the operator to each term:

$$\frac{d}{d\lambda} [\nabla L_{\text{replay}}(\Theta^*(\lambda))] + \frac{d}{d\lambda} [\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))] = 0$$

Evaluating the first term via the multivariable chain rule:

$$\frac{d}{d\lambda} [\nabla L_{\text{replay}}(\Theta^*(\lambda))] = \nabla^2 L_{\text{replay}}(\Theta^*(\lambda)) \cdot \frac{d\Theta^*}{d\lambda}$$

Evaluating the second term via the product rule and chain rule:

$$\frac{d}{d\lambda} [\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))] = (1) \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) + \lambda \cdot \left(\nabla^2 L_{\text{current}}(\Theta^*(\lambda)) \cdot \frac{d\Theta^*}{d\lambda} \right)$$

Substituting these expanded terms back into the equation:

$$\left[\nabla^2 L_{\text{replay}}(\Theta^*) \cdot \frac{d\Theta^*}{d\lambda} \right] + \left[\nabla L_{\text{current}}(\Theta^*) + \lambda \cdot \nabla^2 L_{\text{current}}(\Theta^*) \cdot \frac{d\Theta^*}{d\lambda} \right] = 0$$

Grouping the terms that share the path movement $\frac{d\Theta^*}{d\lambda}$:

$$(\nabla^2 L_{\text{replay}}(\Theta^*) + \lambda \cdot \nabla^2 L_{\text{current}}(\Theta^*)) \cdot \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}}(\Theta^*) = 0$$

Recognising the definition of the combined Hessian H_λ inside the parentheses:

$$H_\lambda \cdot \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0$$

Solving algebraically for the path movement $\frac{d\Theta^*}{d\lambda}$:

$$\frac{d\Theta^*}{d\lambda} = -H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) \quad (\text{D.3})$$

Step 3: Calculating the Change in the Replay Loss. How does the replay loss change as λ increases along this path? We apply the derivative operator:

$$\frac{d}{d\lambda} [L_{\text{replay}}(\Theta^*(\lambda))] = \nabla L_{\text{replay}}(\Theta^*(\lambda))^\top \cdot \frac{d\Theta^*(\lambda)}{d\lambda}$$

We perform two substitutions here. First, from (D.2), we know $\nabla L_{\text{replay}} = -\lambda \cdot \nabla L_{\text{current}}$. Second, from (D.3), we substitute the path movement.

$$\begin{aligned} \frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) &= (-\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)))^\top \cdot (-H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))) \\ &= \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))^\top \cdot H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) \end{aligned}$$

Because H_λ is strictly positive-definite along the followed branch of local minima for all $\lambda \in [0, 1]$ (per the Local-Minimum Assumption), its inverse H_λ^{-1} is also positive-definite. The quadratic form $v^\top M v$ is therefore guaranteed to be non-negative. Since $\lambda \geq 0$, the entire equation satisfies:

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0$$

Conclusion for Part A: The replay loss never decreases along the reference path, so it can never overshoot its final joint-optimum value.

Part B: The Iterate Tracks the Reference Path with $O(1/N)$ Lag

Let $\theta(t)$ be the actual position of the model parameters at time t . We define the tracking error $e(t)$ (written e to avoid a clash with the damping δ of asymmetric PER):

$$e(t) = \theta(t) - \Theta^*(\lambda(t))$$

Step 1: Differentiating the Error. We apply the time derivative operator $\frac{d}{dt}$ to our error definition:

$$\frac{d}{dt} [e(t)] = \frac{d}{dt} [\theta(t)] - \frac{d}{dt} [\Theta^*(\lambda(t))]$$

The model moves via gradient descent, so $\frac{d\theta}{dt} = -\nabla L_{\lambda(t)}(\theta(t))$. The reference path moves via the chain rule $\frac{d\Theta^*}{dt} = \frac{d\Theta^*}{d\lambda} \cdot \frac{d\lambda}{dt}$. Because the curriculum has a constant velocity over N steps, $\frac{d\lambda}{dt} = \frac{1}{N}$.

$$\frac{de}{dt} = -\nabla L_{\lambda(t)}(\theta(t)) - \left(\frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \right)$$

Step 2: Taylor Expansion of the Actual Gradient. We approximate the gradient at the actual position $\theta(t)$ by Taylor-expanding it around the reference position Θ^* :

$$\nabla L_\lambda(\theta) \approx \nabla L_\lambda(\Theta^*) + \nabla [\nabla L_\lambda(\Theta^*)] \cdot (\theta - \Theta^*)$$

Since the gradient at the minimum is zero ($\nabla L_\lambda(\Theta^*) = 0$) and the derivative of the gradient is the Hessian ($\nabla^2 L_\lambda = H_\lambda$), this collapses to:

$$\nabla L_\lambda(\theta) \approx H_\lambda \cdot e$$

Step 3: The Final Steady State. Substitute this expanded gradient back into our differential equation:

$$\frac{de}{dt} \approx -H_\lambda \cdot e - \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N}$$

Assuming the lag reaches a steady state, the error stops growing ($\frac{de}{dt} \approx 0$). This is a first-order, gradient-flow approximation: it linearises the gradient around Θ^* and treats the dynamics as continuous, so it captures the scaling of the lag with N rather than an exact discrete-time bound.

$$\begin{aligned} 0 &\approx -H_\lambda \cdot e - \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \\ H_\lambda \cdot e &\approx -\frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \\ e &\approx -H_\lambda^{-1} \cdot \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \end{aligned}$$

Taking the norm, we see the lag distance is bounded by the curriculum length N :

$$\|e\| = O\left(\frac{1}{N}\right) \tag{D.4}$$

Combining Part A and B: Expanding the Actual Loss

To find the maximum replay loss the model will actually experience, we Taylor expand the loss of the actual trajectory around the reference position:

$$\begin{aligned} L_{\text{replay}}(\theta) &\approx L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top \cdot (\theta - \Theta^*) \\ L_{\text{replay}}(\theta) &\approx L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top \cdot e \end{aligned}$$

From Part A, $L_{\text{replay}}(\Theta^*)$ cannot exceed the final joint optimum loss $L_{\text{replay}}(\theta_{\text{joint}}^*)$. From Part B, the lag e is $O(1/N)$. Substituting these limits yields the final bound:

$$\max_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*) + O\left(\frac{1}{N}\right) \tag{D.5}$$

Conclusion: The transient stability gap is entirely contained within the $O(1/N)$ term, so a sufficiently slow curriculum makes the tracking lag's contribution to the transient arbitrarily small.